

Epitaxial Ge/III-V Heterostructures: MOCVD Growth, Characterization, and Applications

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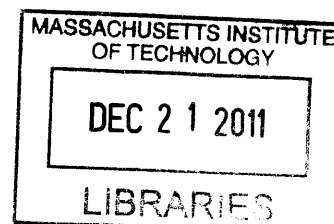
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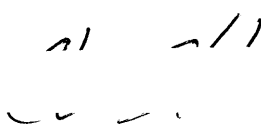
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Abstract

Epitaxial Ge thin films are being investigated for many important roles in next generation microelectronics. Metal-oxide-semiconductor field effect transistors (MOSFETs) utilizing Ge channels have demonstrated dramatic performance enhancements over Si based technology. Theoretical studies have predicted that tensile strain enhances both electron and hole mobilities of Ge to levels much greater than those in unstrained or compressively-strained Ge and Si. Additionally, high enough levels of tensile strain have been postulated to alter the band structure of Ge to make Ge a direct-band-gap semiconductor.

We investigated the physics and fabrication of Ge/III-V compound heterostructures, where the III-V compound material could serve as a tensile-strain-inducing template for subsequent Ge epitaxy. Through experimentation and the use of characterization techniques such as transmission electron microscopy (TEM), photoluminescence (PL), secondary ion mass spectrometry (SIMS) and spreading resistance analysis (SRA), we established correlations between the initial III-V compound surface stoichiometry and the structural, optical, and electronic properties of Ge thin films deposited on GaAs and AlAs templates. We determined that the highest structural quality Ge epitaxy initiates via a bond and exchange mechanism with the surface group III element atoms, whereas group V element atoms do not bond as effectively with Ge and thin film deposition processes that rely on Ge-group V binding processes lead to ‘pitting’ in Ge thin films. With the developed understanding of the growth mechanisms, we successfully fabricated high quality tensile-strained Ge thin films and quantum dots on $\text{In}_x\text{Ga}_{1-x}\text{As}$ templates. Tensile strain levels as high as 0.58% in Ge thin films and 1.37% in Ge quantum dots were achieved. However, the film deposition methods that facilitated the highest structural quality also led to unintentional doping characteristics that affected the electrical and optical properties of the tensile-strained Ge epitaxial structures. Nevertheless, we designed processing sequences that led to the first demonstration of room temperature, Ge direct band gap luminescence from Ge/III-V compound heterostructures.

We parlayed our advancements in Ge/III-V compound heterostructure fabrication to demonstrate a novel process for the fabrication of GaAs-on-Insulator (GaAsOI). The combination of GaAs/Ge/GaAs heterostructure establishment, room temperature oxide-

oxide bonding methods, and XeF_2 -based sacrificial etching of Ge, led to the successful fabrication of GaAsOI on a small scale. We tested the implementation of our process on a full wafer scale and determined the process was kinetically limited by the lateral Ge etch process. We adapted the Deal-Grove oxidation model to establish a model to understand the relationships between lateral etch rate, lateral etch distance, release layer thickness, channel displacements, and the radius of curvature of the donor wafer.

Our Ge/III-V compound heterostructure research advanced the understanding and state-of-the-art processing of such structures. We established methods and elucidated challenges for future research targeted toward demonstrating novel Ge-based devices and advanced large-diameter engineered substrates.

Thesis Supervisor: Eugene A. Fitzgerald

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Chapter 1 Introduction

1.1 Background and motivation for investigation of Ge/III-V heterostructures

1.1.1 Strain engineering and advanced CMOS channels

As Si-based CMOS technology approaches the end of its scaling road map, the industry has been exploring new materials for devices with better performance to continue serving the ever-growing demand for computing power. In the last decade, strained-Si technology has been used to enhance the performance of metal-oxide-semiconductor field effect transistors (MOSFETs)¹⁻⁴. To further improve the MOSFET performance, researchers have continued to explore new materials with higher intrinsic carrier mobilities. Among these materials, Ge has been actively studied due to its intrinsically high hole mobility. To date, most of the MOSFETs utilizing Ge channels have been built on bulk Ge substrates^{5,6} or from compressively strained Ge thin films epitaxially grown on SiGe virtual substrates⁷⁻⁹. Compressive strain enhances Ge hole mobility but degrades electron mobility, making it suitable for p-MOSFETs but disadvantageous for n-MOSFETs. It has been demonstrated that compressively strained Ge channels exhibit 10-12× hole mobility enhancement over Si channels⁸. On the other hand, tensile-strained Ge has not been very well studied due to the difficulty of strain engineering with tensile Ge thin films. In theory, tensile strain enhances both electron and hole mobilities of Ge to levels much greater than those in unstrained or compressively-strained Ge and Si¹⁰, making it a highly promising channel material for future CMOS applications. There is a need for growth techniques to produce high quality tensile-strained Ge with different strain levels to fully explore these theoretical predictions. III-V compound semiconductors such as InGaAs

and InAlAs offer a full range of lattice constants up to 4% larger than Ge lattice constant, and thus can serve as an ideal strain-inducing epitaxial template. To produce highly tensile-strained Ge layers, heteroepitaxy of Ge on III-V materials with larger lattice constants is one of the most feasible methods.

1.1.2 Ge for novel optical devices

In addition to enhanced carrier mobilities, The difference in energy level between the direct (Γ_c - Γ_v) and indirect (L_c - Γ_v) band gap of Ge is small at room temperature (~ 0.13 eV) and biaxial tensile strain in (100) Ge is predicted to reduce this difference, making Ge an “almost-direct-bandgap material”¹⁰⁻¹² with potential to enable novel applications in optoelectronic devices such as better performance photodetectors, laser diodes and modulators. Performance enhanced tensile Ge photodetectors integrated on Si have been demonstrated¹³ and light emitting devices with Ge as the gain medium are being researched^{14,15}. In 2009, Sun *et al.* reported Ge band engineering with Ge grown on Si substrates and how tensile strain and n-type doping can be applied to significantly enhance the direct band gap optical transition in ref [16]. Based on these results, Liu *et al.* demonstrated optical gain and optically pumped lasing in Ge-on-Si in subsequent studies^{14,15} Computational studies have predicted that biaxial tensile strain greater than 1.4% effectively turns Ge thin films into a direct band gap material. III-V compounds as strain-inducing templates are capable of applying large tensile strains in Ge epitaxial thin films. Ge/III-V heterostructures offer us a potential platform to explore the optical properties of tensile-strained Ge for novel optical devices.

1.1.3 Monolithic III-V integration on Si and engineered substrates

Monolithic integration of silicon CMOS circuits with III-V optical and electronic devices is another frontier of microelectronics research. Intensive investigation has been conducted on MOSFETs built with III-V semiconductor materials like GaAs and InGaAs due to their high intrinsic carrier mobility¹⁷⁻²⁰. III-V-based optical interconnects present a potential solution to reduce RC delay, another bottle neck for high speed Silicon ICs. In the last decade, integration of III-V based optical devices has become one of the most studied challenges of modern age microelectronics.²¹⁻²⁴ Beyond Si CMOS, integration of III-V based monolithic microwave integrated circuits (MMICs) with Si base logic circuits on a signal die will eventually raise the scale and depth of integration to a new level, and boast the performance and functionality of future generation integrated circuits²⁵⁻²⁷.

Development in engineered substrates has enabled key advancements in these areas. For over a decade, Si-on-insulator (SOI) technology has been used in mass production of integrated circuits to reduce transistor leakage current and enhance overall circuit performance. The development of Ge channel MOSFETs and successful demonstrations of monolithically integrated III-V compound/Si circuits have promoted the development of Ge-on-insulator (GeOI) substrates and Si-on-lattice-engineered-substrates (SOLES).^{28,29} InAlAs/InGaAs based high electron mobility transistors (HEMTs) and InGaAs/InP based heterojunction bipolar transistors (HBTs) have been fabricated on GeOI substrates and SOLES.^{25,26}

By far, Ge has played an essential rule in enabling these engineered substrates. Ge has nearly the identical lattice constant as GaAs, the most commonly studied and employed III-V materials. As a group IV semiconductor with properties similar to Si and the ability

to be metallurgically alloyed with Si at any content, Ge has long been integrated with Si to make strained channel devices and HBTs. With its unique structural properties, Ge naturally serves as the best media to bridge III-V and Si materials. Advanced engineered substrates such as, relaxed SiGe substrates (also known as ‘metamorphic’ SiGe), GOI substrate and SOLES were all enabled through Ge epitaxial structures. The future of III-V integration on Si requires more variety of engineered substrates, such as low cost GOI and GaAsOI. High quality Ge/III-V heterostructures have the potential to enable novel fabrication process for low cost manufacturing of advanced engineered substrates.

1.2 Influence of biaxial tensile strain on electronic and optical properties of (100) Ge

The unique electronic and optical properties of Ge make it the desired candidate in advancing the above mentioned fields. This section provides background on how these properties originate from the unique band structure of Ge, and the important role biaxial tensile strain in enhancing these desired properties.

Computational results have predicted that in-plane mobility for both electrons and holes in (100) Ge thin films increase as biaxial tensile strain is applied to the film, illustrated in Figure 1.1a. In addition, high levels of biaxial tensile strain can convert Ge into a direct band gap semiconductor, as shown in Figure 1.1b¹⁰. These effects originate from the changes in Ge band structure under biaxial tensile strain, illustrated in Figure 1.2. Biaxial tensile strain, denoted by ε in the figure, splits the energy levels of the light hole and heavy hole bands at the valence band maxima, shifting holes to the light hole band, resulting in enhanced hole mobility.

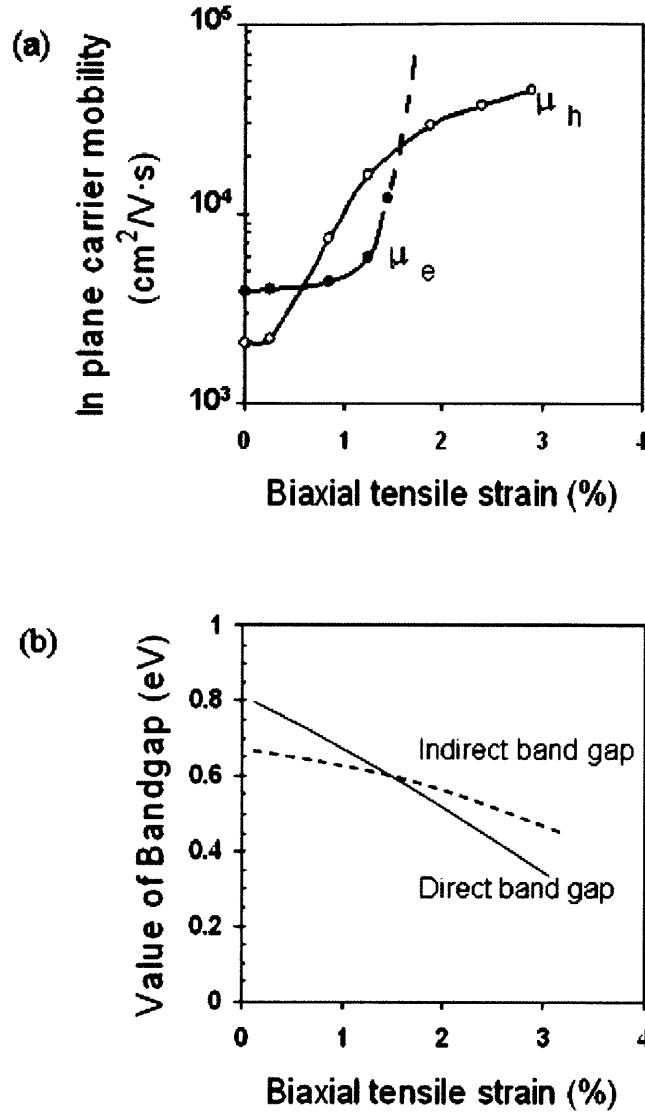


Figure 1.1. Calculated carrier mobilities (a) and bandgap energy (b) of (100) Ge as function of biaxial tensile strain (Figures reproduced from Ref [10]).

On the conduction band side, as the direct band gap E_Γ shrinks faster than the indirect band gap E_L with biaxial tensile strain, the probability of electrons populating the Γ valley increases. Because the Γ valley has a lower in-plane effective mass than the L valley, the collective electron mobility increases slowly as more electrons populate the Γ valley. When the biaxial tensile strain is increased above about 1.4% strain, the Γ valley is lower than the L valley, $E_{\Gamma,\epsilon} < E_{L,\epsilon}$, as shown in Figure 1.1b, making Ge a direct band gap

material. In addition, most electrons in the conduction band will now reside in the Γ valley. This explains the abrupt increase in electron mobility at about 1.4% tensile strain in Figure 1.1a. Furthermore, as the direct band gap shrinks, the curvatures of the light hole band and the conduction band Γ valley increase, resulting in further enhancement of hole and electron mobilities.

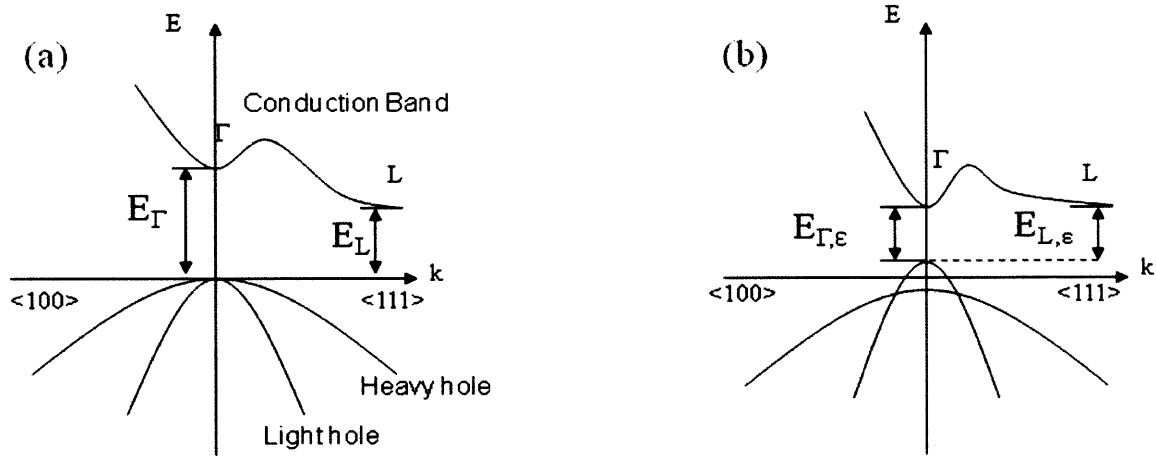


Figure 1.2. Ge band structure without (a) and with (b) ~1.4% biaxial tensile strain.

In fact, tensile-strained Ge potentially possesses some of the highest electron and hole mobilities among group-IV and III-V materials (Table 1.1). Furthermore, unlike Si or GaAs, the electron and hole mobilities can be comparable in tensile-strained Ge at a certain amount of strain, making it an excellent material for both *p*-MOS and *n*-MOS.

	Carrier mobility (cm ² /V·s)	
	Electron	Hole
1.5% tensile-strained Ge	~12000	~20000
Ge	4000	2000
Si	1400	450
GaAs	8500	400

Table 1.1. Comparison of carrier mobilities in tensile-strained Ge with other semiconductors under consideration for high speed CMOS channels

When the Ge direct band gap and indirect band gap crossover at about 1.4% of tensile strain, the direct band gap is predicted to have a value about 0.6 eV, corresponding to about 2 μm in band-to-band emission wavelength. Under even higher tensile strain, the band-to-band emission wavelength can be as long as 10 μm .³⁰ On the other hand, considering quantum confinement in device design, it is also feasible to tune the emission wavelength down to about 1.5 μm for tensile-strained Ge quantum wells. The potential of covering a wide range of emission wavelength makes tensile-strained Ge an excellent candidate for optoelectronic devices emitting in the near-IR range.

We pursue the growth and properties of Ge/III-V epitaxial heterostructures within this thesis. One goal is to induce a large amount of tensile strained to (100) Ge thin films, therefore allowing us to test the theoretical predictions and possibly open the path for application in novel devices and circuits.

1.3 Organization of this thesis

Chapter 2 details the materials growth and characterization techniques that were employed for the work described in the thesis. Chapter 3 presents the set of experiments of Ge growth on GaAs and detailed discussion on the impact of heterovalency during Ge epitaxy on GaAs surfaces independent of strain. Chapter 4 follows with a photoluminescence study of these Ge epitaxial thin films on GaAs and correlates the luminescence properties with the unintentional doping of Ge by Ga and As during the epitaxy. These results help us further our understanding of the initiation process of Ge

epitaxial growth on GaAs. In Chapter 5 we extend the result from Chapter 3 and 4 and demonstrate that III-V compound such as $\text{In}_x\text{Ga}_{1-x}\text{As}$ can indeed be used as epitaxial template to induce large amount of strain in (100) Ge. In Chapter 6, we present growth of more complex Ge/III-V heterostructures and discuss how they can be used to fabricate advanced engineered substrates. Chapter 7 presents a novel low cost process for fabrication of GaAsOI structures. We present the successful demonstration of this process on small scale and discuss the key considerations to apply this process to fabrication of large diameter GaAsOI substrates. Finally, in Chapter 8, we summarize the work in this thesis, and offer suggestions of future paths that can be explored to take our work further.

Chapter 2 Materials Growth and Characterization

2.1 Metalorganic chemical vapor deposition (MOCVD)

All the heterostructures presented in this study were grown using a custom-built Thomas Swan/AIXTRON low pressure MOCVD system with a close-coupled showerhead configuration. An image of the reaction chamber is shown in Figure 2.1

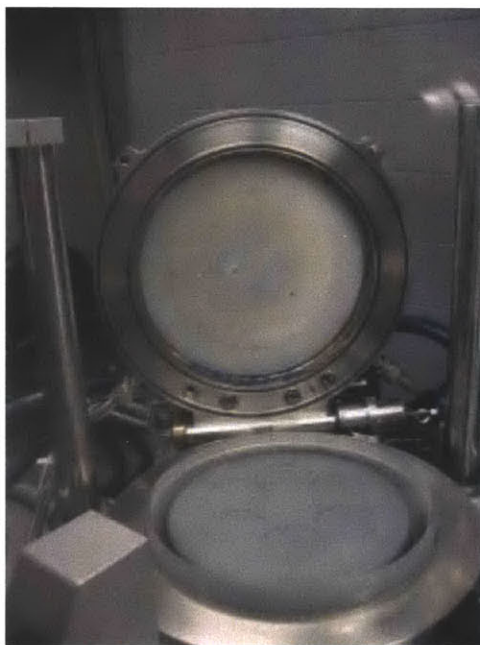


Figure 2.1. Photograph of the reaction chamber of the MOCVD that was used to grow the samples reported in this thesis. The flip-top showerhead, through which all gases are injected into the chamber, is currently raised, revealing the 6 × 2" susceptor is currently installed in the chamber.

Like most MOCVD reactors set up to grow As- and P- films, our reactor is installed with the group-III metalorganic sources trimethylaluminum (TMAl), trimethylgallium (TMGa) and trimethylindium (TMIn) which serve as the aluminum (Al), gallium (Ga) and indium (In) precursors respectively, as well as the group-V hydride sources arsine (AsH_3) and phosphine (PH_3) which are the precursors for As and P. Dimethylzinc (DMZn) and 0.01% (balance H_2) disilane (Si_2H_6) are the p-type and n-type dopant sources, respectively. In addition, the reactor is also installed with Isopropyl alcohol (IPA) as an

oxygen source for growing Al_2O_3 ,³¹⁻³³ as well as trimethylantimony (TMSb) for growing Sb-films.

What truly sets our system apart from most other MOCVDs in operation is the added presence of group-IV sources, specifically silane (SiH_4) and germane (GeH_4), as well as 1% diborane (B_2H_6) for p-doping (n-doping is carried out using PH_3 , though AsH_3 can potentially be used as well). This means that the reactor has the unique capability to fabricate Si, Ge, and III-V thin films, thus allowing for the *in situ* grown Ge/III-V heterostructures demonstrated in this study. Our MOCVD reactor was the first ever system with such a capability when it was commissioned in 2005, and, to the best of our knowledge, is likely still the only such system in the world. More details of the operation of this system can be found in Lee's thesis.³⁴

The system uses Nitrogen (N_2) and hydrogen (H_2) as carrier gas. For all growth Nitrogen was used as the carrier gas and the chamber pressure was 100 Torr. The growth temperatures were established by optical pyrometry measurements of the wafer surface.

The system is also equipped with an EpiTT that monitors the reflectivity of the growth surface at wavelength of 635 nm in real time, providing information about growth transitions, initiation times, and layer thicknesses. To illustrate, Figure 2.2 shows the wafer surface reflectivity as a function of growth time for a Ge growth on GaAs at 475 °C. Before Ge growth started, the GaAs surface reflectivity was constant at 0.35. At 101 sec,

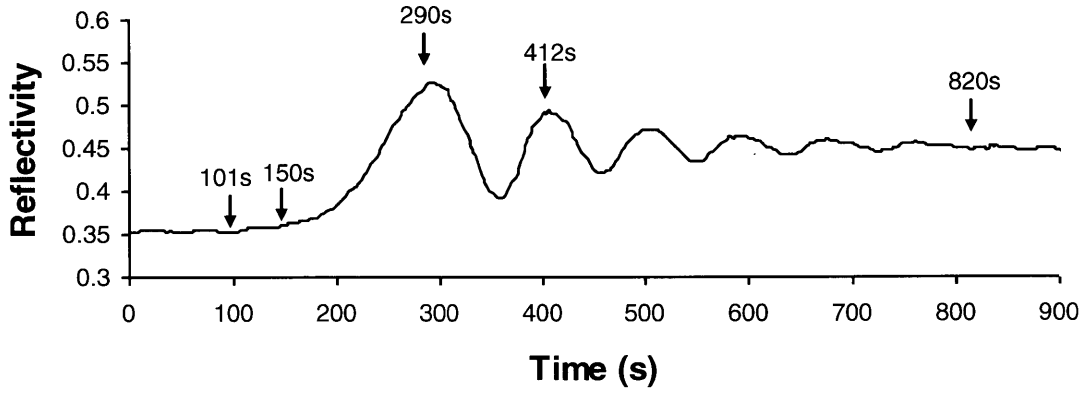


Figure 2.2. Reflectivity of wafer surface recorded by EpiTT as a function of growth time for a growth of Ge on GaAs at 475 °C; $t=101$ s, GeH_4 turned on; $t\sim 105$, incubation period; $t=290$ s, first peak of constructive interference; $t=412$ s, second peak of constructive interference; $t=820$ s, GeH_4 turned off.

GeH_4 was introduced to the chamber. Since Ge is more reflective than GaAs at 635 nm wavelength, the surface reflectivity increases upon Ge deposition. During the growth, the reflectivity peaks when constructive interference occurs between the reflected beams from the Ge and GaAs surfaces. This condition occurs when $2d=m\lambda/n$ is satisfied, where d is thickness of the Ge film; m is any natural number; $\lambda=635\text{nm}$ is the wavelength of light source used by EpiTT; and n is the Ge refractive index for $\lambda=635\text{nm}$ at 475 °C, estimated to be 5.59 from ref [35,36]. The valleys of reflectivity intensity occur when $2d=(m+1/2)\lambda/n$ is satisfied. The film thickness can be estimated by counting the peaks. For example, at 290 sec, m equals 1 and the Ge film thickness d_1 is estimated to be 57 nm. At 414 seconds, the Ge thickness is about 114 nm, $d_2=2d_1$. It can be observed that the time needed to grow the first d_1 (~190 seconds) is longer than that needed for the second d_1 (~120 seconds). This suggests that there is an incubation period at the beginning of the Ge growth on GaAs, which we examined in this study.

2.2 Characterization techniques

The composition and strain of the epitaxial thin films were characterized by x-ray diffraction (XRD). Strain levels of Ge thin films were measured by Raman spectroscopy using a Kaiser Hololab system with a 514nm Ar-ion laser in a backscattering configuration. The surface morphology of the Ge thin films was characterized by atomic force microscopy (AFM). Cross-sectional transmission electron microscopy (XTEM) was used to accurately determine the film thickness and examine the film quality. XTEM specimens were prepared with traditional mechanical grinding techniques, following by an argon ion milling step to establish regions that were electron transparent. The XTEM imaging was conducted on a JEOL 2010FX system. Photoluminescence (PL) spectra of the thin film samples were taken in a backscattering configuration using a 514-nm Ar ion laser with an excitation intensity of 20 W/cm². An InGaAs photodiode with response range of 900-1700 nm was used as the detector to obtain the spectra. The sample stage was cooled with compressed liquid He to reach a minimum temperature of 20 K. Secondary ion mass spectrometry (SIMS) and spreading resistance analysis (SRA) were used to analyze the impurity and active carrier concentrations in the thin films. SRA was conducted by Solecon Labs, Inc and SIMS analysis was provided by Evans Analytical Group.

Chapter 3 Growth Initiation of Ge on GaAs

3.1 Growth initiation of Ge on GaAs

Unlike epitaxy in Ge/SiGe/Si or Ge/GeSn/Si systems, where all the layers consist of only non-polar group IV elemental semiconductors, Ge epitaxy on III-V compound semiconductors involves a heterovalent interface of between polar and non-polar semiconductors. A few groups have investigated Ge growth on GaAs substrates using MBE systems but reported different results.³⁷⁻⁴⁰ A few of them showed non-planar growth even though the lattice mismatch between Ge and GaAs is negligibly small^{37,38}. This was attributed to the difference in valence structure between Ge and GaAs. The even fewer reports on Ge growth on GaAs via CVD did not provide details of the growth process and its influence on film morphology.^{41,42} Our first challenge was to understand the impact of heterovalency on Ge epitaxy and to identify a growth technique for high quality Ge epitaxy on III-V substrate with low-pressure MOCVD.

The different valence characteristics at the GaAs (100) surface pose a challenge to Ge epitaxy. Substrate surfaces greatly affect the initiation of epitaxial growth and thus determine the morphology and the quality of the films. The GaAs surface is known to have different surface reconstructions at different temperatures or under different ambient conditions⁴³. Figure 3.1 shows the ball-and-stick model of two surface reconstructions with very different surface As or Ga coverage. Table 3.1 summarizes the possible surface reconstructions at different temperature range and ambient conditions. These surface reconstructions have different stoichiometry and thus different valence structures^{44,45}, which can potentially affect Ge growth initiation.

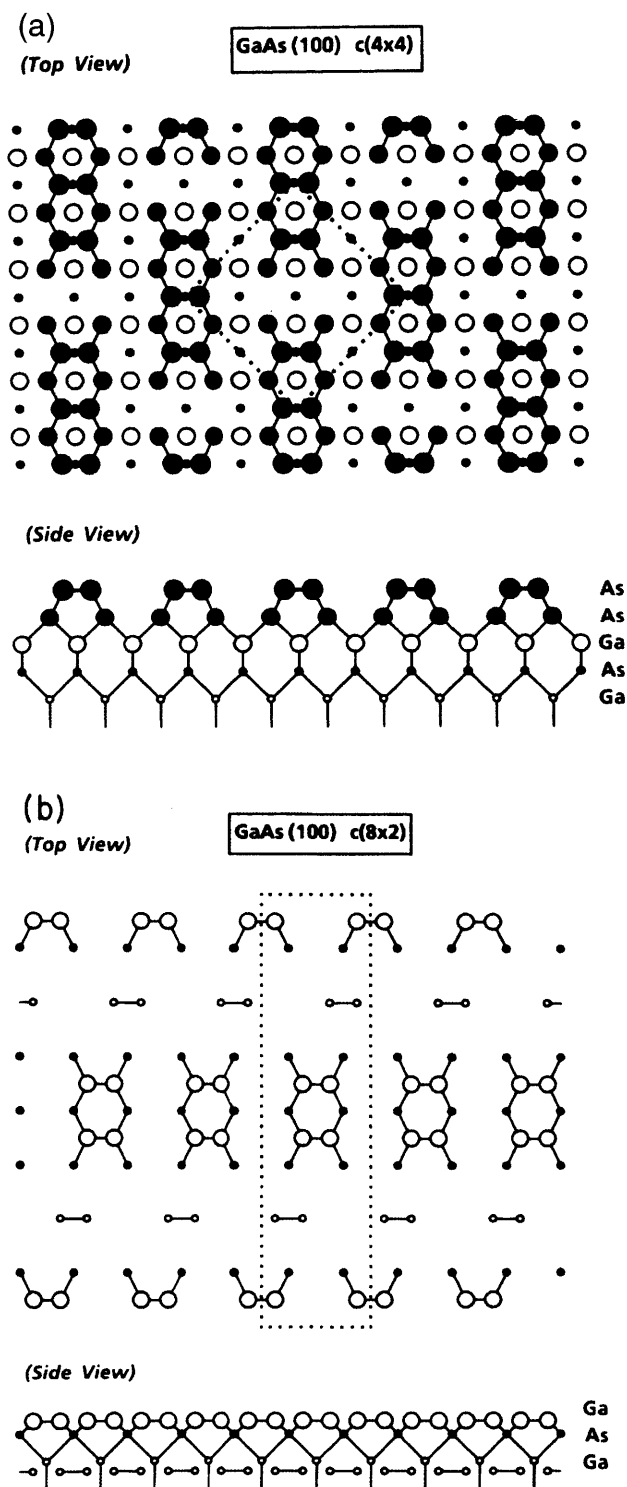


Figure 3.1. Ball-and-stick model of two possible reconstructions of (100) GaAs surface: a) As-rich $c(4 \times 4)$ reconstruction, b) Ga-rich $c(8 \times 2)$ reconstruction, after biegersen et al.⁴³

In order to study this effect, we altered the stoichiometry (Ga-to-As ratio) of the GaAs surfaces by employing different surface preparation techniques and compared the quality of Ge epi-layers initiated on these different surfaces. Figure 3.2 shows the morphology of Ge thin films grown on GaAs substrates with three different surface preparation techniques. Table 3.2 summaries the growth procedure for each of them and two other samples not shown in Figure 3.2, which will be discussed in later in this section.

Table 3.1. List of surface reconstructions of (100) GaAs under different temperature range and preparation mode, and their corresponding surface As coverage ratio, after Massies et al.⁴⁴

Temperature range (°C)	Surface structure	Preparation mode	Surface coverage θ_{As} (after ref. [13])
650	c(8 × 2)	MBE or annealing of c(2 × 8)	0.22
600–550	(4 × 1)		
600–550	(3 × 1)		
600–550	c(6 × 4)		0.37
500–450	(1 × 6)		0.52
500–450	c(2 × 8)	As adsorption or annealing of (1 × 1)	0.61
350–300	c(4 × 4)		0.86
150	(1 × 1)		?

Table 3.2. Process variants of different surface preparation techniques to initiate Ge on GaAs (100) surface

Growth Steps	Sample <i>a</i>	Sample <i>b</i>	Sample <i>c</i>	Sample <i>d</i>	Sample <i>e</i>
1	GaAs homoepitaxy at 650 °C				
2	Cooling to 475 °C under 100 sccm AsH ₃ flow	Cooling to 475 °C under 100 sccm AsH ₃ flow	Anneal at 650 °C under N ₂ ambient for 10 minutes	Cooling to 350 °C under 100 sccm AsH ₃ flow	Cooling to 475 °C under 100 sccm AsH ₃ flow
3	Shut off AsH ₃ , pause for 10 seconds	Shut off AsH ₃ , pause for 10 seconds	Cooling to 350 °C under N ₂ ambient	Shut off AsH ₃ , pause for 10 seconds	Shut off AsH ₃ , pause for 10 seconds
4	Ge growth at 475 °C with 100 sccm GeH ₄ flow	Flow 50 sccm TMGa for 15 seconds	Ge growth at 350 °C with 100 sccm GeH ₄ flow	Flow 50 sccm TMGa for 15 seconds	Flow 50 sccm TMGa for 15 seconds
5		Shut off TMGa, pause for 10 seconds		Shut off TMGa, pause for 10 seconds	Cooling to 350 °C under N ₂ ambient
6		Ge growth at 475 °C with 100 sccm GeH ₄ flow		Ge growth at 350 °C with 100 sccm GeH ₄ flow	Ge growth at 350 °C with 100 sccm GeH ₄ flow

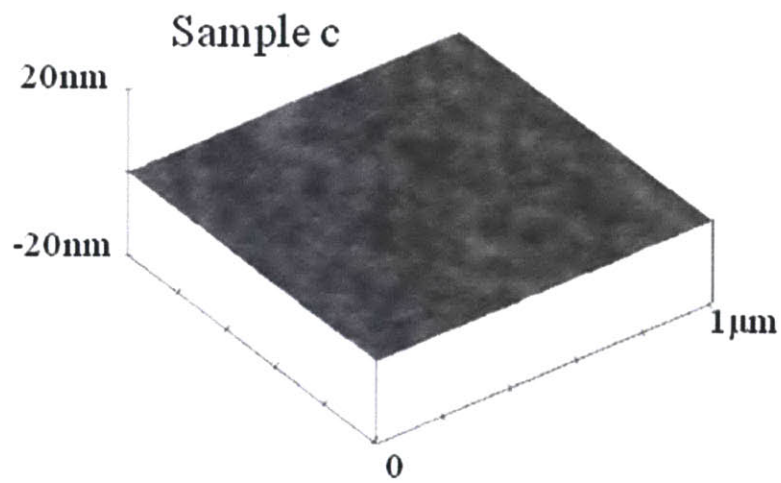
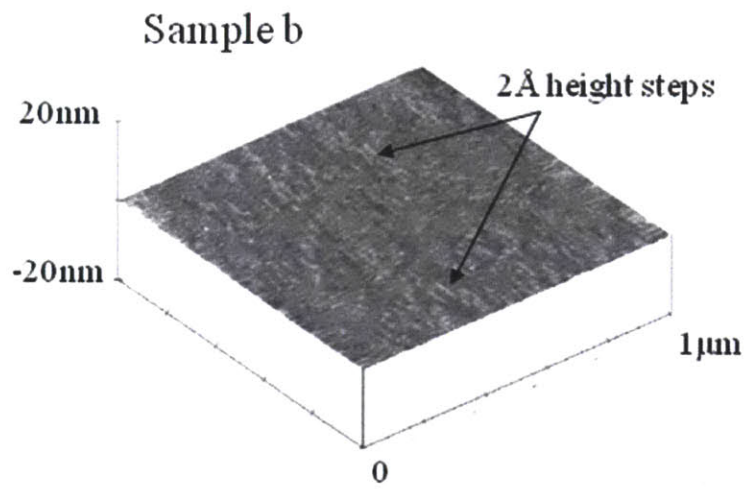
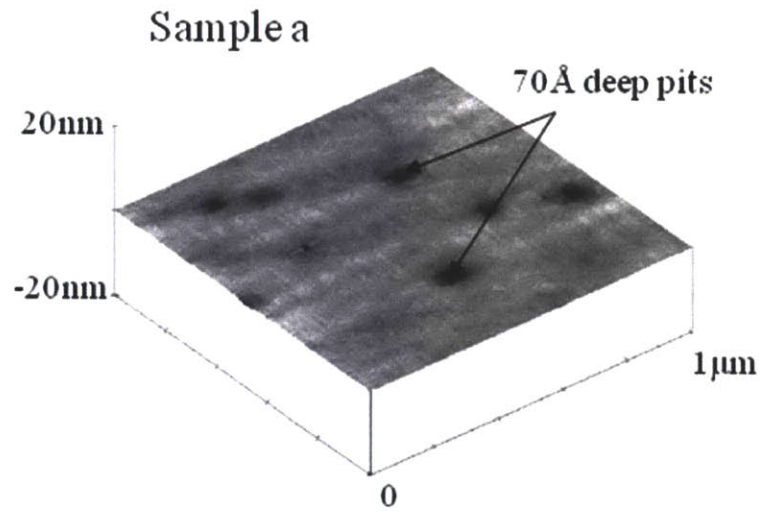


Figure 3.2. AFM images of Ge initiation layers grown on (100) GaAs surface with low Ga-to-As ratio (sample a) and high Ga-to-As ratio (sample b and c).

For all samples, (100) GaAs substrates were used and a GaAs homoepitaxial layer was grown at 650 °C to ensure a high quality epi-ready surface. For sample *a*, an AsH₃ overpressure was kept until 10 seconds before Ge growth. This procedure preserves the typical GaAs surface obtained during GaAs MOCVD, which is known to be a c(4×4) or d(4×4) surface reconstruction as illustrated in Figure 3.1a, with multiple layers of As on the surface ⁴⁶. For sample *b*, a 15 second pulse of TMGa flow was inserted between shutting off the AsH₃ flow and turning on the GeH₄ to consume the excess As and increase the Ga-to-As ratio at the surface. For sample *c*, instead of flowing TMGa to consume the excess As on the surface, the substrate was annealed in N₂ ambient for 10 minutes at 650 °C before cooling down to Ge growth temperature. Given that As has a high vapor pressure at this temperature, the anneal should drive some of the As off the surface and form surface reconstructions with higher Ga-to-As ratio (*e.g.*, (4×2) or (8×2) reconstruction). The substrate was then cooled down under an N₂-only ambient to the Ge growth temperature and Ge was directly initiated on this surface. Due to different surface conditions and different Ge growth temperatures, it was difficult to keep the Ge thickness identical for all three samples. However, the Ge layer thickness was kept low enough (on the order of 10 nm) to study the film morphology at the initial stage.

As shown in Figure 3.2, the AFM image reveals pits in the Ge layer in sample *a*. In the XTEM image shown in Figure 3.3a, the pits appeared to penetrate the entire film thickness. The formation of the pits destroys the integrity of the film and prevents the possibility of FET device fabrication with this Ge layer. For sample *b*, the surface is flat. The features on the surface have average step height of 2 Å, corresponding to the atomic

steps on the Ge surface. The RMS roughness of this sample is only 0.14 nm, about the lowest that can be achieved during epitaxy. For sample *c*, the surface was as flat as that observed with sample *b*, again with no pit formation, but the atomic steps were not as distinct as in sample *b*. A cross-sectional TEM picture of sample *c* is shown in Figure 3.3b.

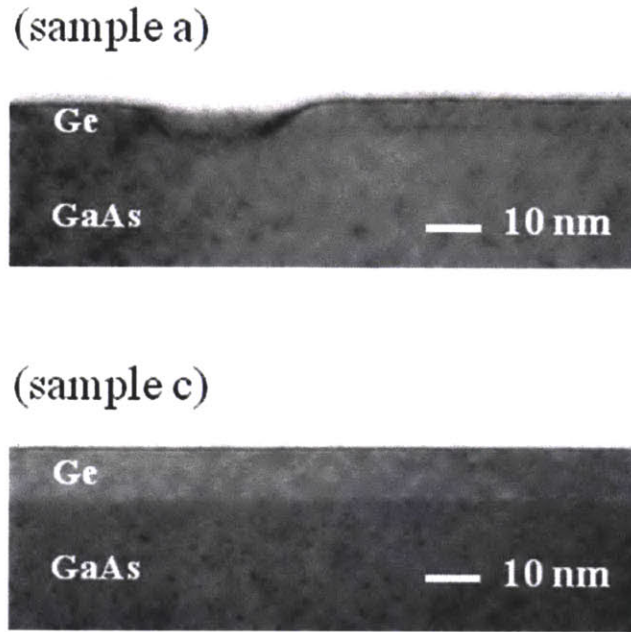


Figure 3.3. XTEM images of Ge initiation layers grown on (100) GaAs surface with low Ga-to-As ratio (sample a) and high Ga-to-As ratio (sample c).

The results indicate that a surface with higher Ga-to-As ratio facilitates high quality Ge epitaxy. The results are consistent for a wide range of Ge growth temperatures, between 350 °C and 500 °C. A possible explanation of this phenomenon relates to the formation energy of Ge-Ga bonds and Ge-As bonds. Wang *et al.* observed that during Ge epitaxy on GaAs by MBE, the Ge film initiated by forming a Ge-Ga dimerized (1×2) surface, which suggests that the Ge-Ga dimer has a lower energy state than the Ge-As dimer³⁷. Our observations are consistent with this theory. For the case of sample *a*, where the

GaAs surface was covered by multiple layers of As, Ge adatoms needed to diffuse through the As layers to bond with Ga atoms. We speculate that the excess As atoms formed clusters that locally impeded Ge adatom attachment, facilitating the pit formation. For the GaAs surfaces with a high Ga-to-As ratio (samples *b* and *c*), there were more available surface sites for Ge adatoms to bond, resulting in a uniform epi-layer. When thicker films were grown using the procedures for samples *b* and *c*, the films were found to retain flatness regardless of the thickness.

3.2 *TMGa pyrolysis on GaAs surface*

We demonstrated two different methods to achieve GaAs surfaces with high Ga-to-As ratio. The method of pulsing TMGa to consume excess surface As works best at higher Ge growth temperature (450 °C - 500 °C). For example, for sample *b* described above, the TMGa pulse was introduced to the chamber at 475 °C, followed by GeH₄ flow at the same temperature and the film was high quality. However, for another sample not shown in Figure 3.2 or Figure 3.3 (sample *d* in Table 3.2), when TMGa was flowed at 350 °C, the subsequent Ge growth at 350 °C had a very long incubation time, as shown by EpiTT data in Figure 3.4 line 3, and yielded a film with much higher surface roughness. We attribute this to the likely incomplete pyrolysis of TMGa at 350 °C. In another sample, when TMGa was flowed at 475 °C followed by cooling the substrate to 350 °C under an N₂-only ambient to initiate Ge growth (sample *e* in Table 3.2), the initiation time was reduced (Figure 3.4 line 2) but was still longer than the case for sample *c* described in Table 3.2 (Figure 3.4 line 1), which was grown at the same temperature with the N₂ anneal at 650 °C surface preparation technique. In addition, sample *e* had surface roughness of 0.27 nm, while sample *c* has lower roughness of 0.18 nm. Furthermore,

SIMS analysis revealed a carbon concentration peak of $3 \times 10^{19} \text{ cm}^{-3}$ at the Ge/GaAs interface in sample *e* (not shown here).

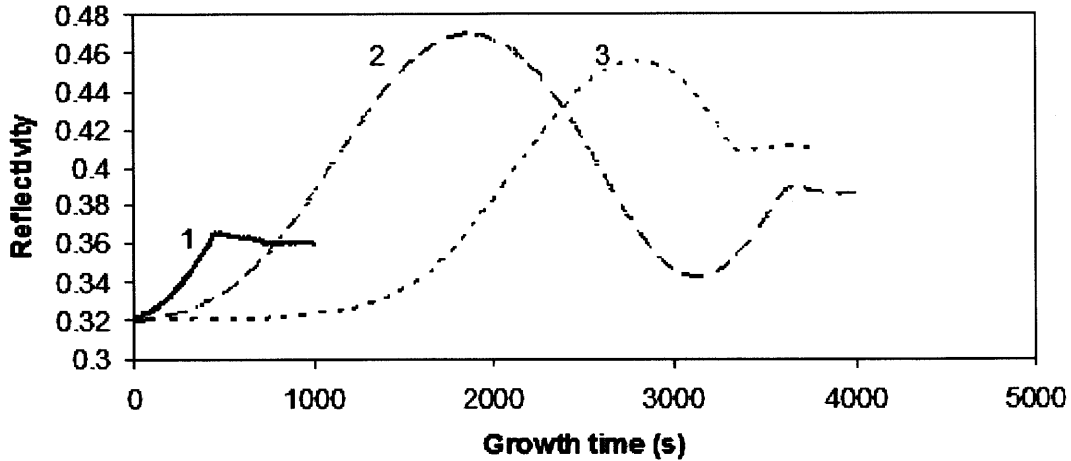
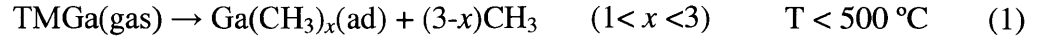


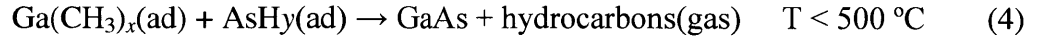
Figure 3.4. Wafer surface reflectivity recorded by EpiTT as a function of growth time: 1) Surface annealed at 650 °C, Ge growth at 350 °C; 2) TMGa flowed at 475 °C, Ge growth at 350 °C; 3) TMGa flowed at 350 °C, Ge growth at 350 °C. Line 1 indicates the shortest initiation time.

To explain the phenomena described above, we need to discuss the details of TMGa pyrolysis and its reaction with the surface. Larsen *et al.* have shown that TMGa pyrolysis is completely inhibited at temperatures below 400 °C in a nitrogen ambient⁴⁷. Thus only a very small fraction of the TMGa introduced at 350 °C reacts with the surface, and the surface Ga-to-As ratio does not change significantly. Most of the TMGa was simply pumped away. When GeH₄ was introduced in the next step, the surface it faced was still covered by excess As, leading to the long incubation time and defective Ge layer.

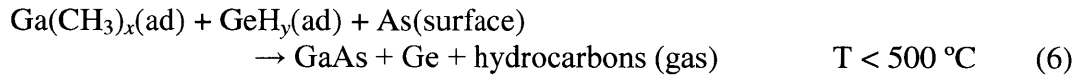
At temperatures above the minimum pyrolysis temperature, it was suggested that TMGa pyrolysis produces gallium complexes in the form of Ga(CH_n)_x. Under normal reaction condition, *n* equals to 3 and the process can be summarized by the following reaction scheme⁴⁸⁻⁵⁰.



For GaAs growth under 500 °C, GaAs is formed through the following reaction:



The temperature 500 °C here is the threshold temperature of complete pyrolysis, but it can be as low as 465 °C for H₂ ambient and as high as around 530 °C for N₂ and He ambient. [Stringfellow p164]^{47,51}. When TMGa was introduced to the chamber of a N₂-only ambient at 475 °C, partial pyrolysis occurred as described by reaction (1). Absence of AsH₃ prevented reactions reaction (3) and (4). The gallium complexes, mainly monomethylgallium(MMGa) and dimethylgallium(DMGa), migrate on the surface until trapped by surface As atoms, forming a self-limiting monolayer. This monolayer absorption of gallium complex is utilized as the self-limiting mechanism for atomic layer epitaxy (ALE) of GaAs^{52,53}. Without AsH₃ in the ambient, the Ga-C bonds in MMGa and DMGa disassociate only when GeH₄ was introduced, through the following reaction:



Therefore the temperature at which GeH₄ is first introduced determines the temperature where reaction (6) can proceed and affects the decomposition of the gallium complexes. It has been found that during low temperature GaAs growth with TMGa as the Ga source, a small portion of the third Ga-C bond would not disassociate, resulting in C incorporation in the GaAs as a p-type dopant⁵⁴⁻⁵⁷. The carbon doping level was found to increase monotonically with decreasing growth temperature and levels on the order of

10^{20} cm^{-3} were obtained through this mechanism at temperatures around 400 °C. This explains the carbon concentration of $3 \times 10^{19} \text{ cm}^{-3}$ found at the Ge/GaAs interface for the sample where GeH_4 was introduced at 350 °C after the TMGa pulse at 475 °C. The long incubation time and high roughness of this sample could be induced by high C concentration or incomplete As consumption on the surface. In contrast, for sample *b* where the TMGa pulse was introduced to chamber at 475 °C and GeH_4 growth was started at the same temperature, the carbon concentration was found to be $2 \times 10^{17} \text{ cm}^{-3}$, and the film was much smoother.

These results show that the method of flowing TMGa is difficult to implement at low growth temperature, around 350 °C. In contrast, the second method to obtain high Ga-to-As ratio, which is simply to anneal the surface at higher temperature (500°C to 650°C) in a N_2 -only ambient before cooling down to the Ge growth temperature, was found to be more robust and reliable and yielded smoother Ge films. Annealing at 500°C to 650°C for 5 to 15 min was effective at yielding high quality Ge growth on GaAs at any temperature between 350°C and 500°C. We applied the annealing preparation method for the rest of our experiments, with slight modifications when applicable.

3.3 Effects of substrate orientation on Ge thin film morphology

Offcut substrates are commonly used for growth of GaAs on Ge, in order to avoid anti-phase boundary (APB) formation in the GaAs layer⁵⁸⁻⁶⁰. APB formation is not a concern for Ge growth on GaAs (APBs cannot form in Ge with only one element in the layer), and as shown above, high quality Ge thin films were grown on exact-(100) GaAs

substrates. Nevertheless, fabrication of Ge optoelectronic devices may require Ge layers to be sandwiched between two III-V semiconductor layers to obtain desired quantum confinement characteristics. Offcut substrates would be preferable for these structures to suppress APB formation in the III-V capping layer. It has been shown that a 6° offcut is sufficient to eliminate APBs in GaAs grown on Ge. Hence, we grew Ge on a (100) GaAs substrate with 6° offcut towards the [011] direction to investigate the influence of substrate offcut on Ge growth.

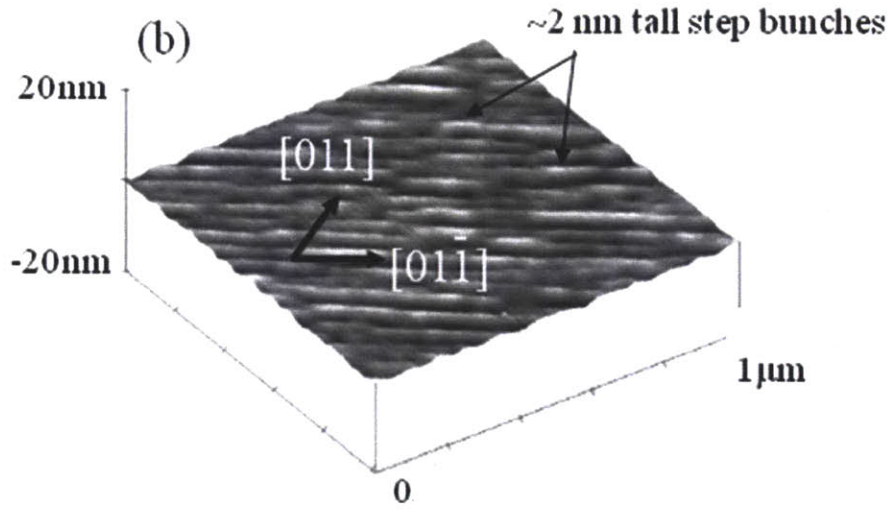


Figure 3.5. AFM images of Ge grown at 475°C on (100) GaAs substrate with 6° offcut towards [011], using optimum surface preparation for Ge initiation.

Figure 3.5 shows the surface morphology of a Ge thin film grown on this offcut substrate. Using the same surface preparation and growth condition for sample *b* as described in Table 3.2, the Ge film grown on this 6° offcut substrate was found to form step bunches along the $[01\bar{1}]$ direction, equally spaced in the [011] direction. The height of these step bunches are about 2 nm and the spacing between them is about 50 nm, resulting in a surface RMS roughness of 0.64 nm, about four times higher than the Ge film grown on

substrate without offcut under the same condition. The step bunches and extra roughness caused by using offcut substrates might pose a challenge to future device fabrication.

3.4 Summary

In this chapter, we have observed that surfaces with high Ga-to-As ratio are essential to initiate smooth, pit-free Ge thin films on GaAs. We have investigated different surface preparation methods and found that annealing under N₂-only ambient between 550°C and 650°C is a reliable method to achieve surfaces with high Ga-to-As ratio. With these procedures, we have grown structurally high quality Ge thin films on GaAs. In the next chapter, we will investigate the optical properties of these Ge thin films.

Chapter 4 Photoluminescence and Electronic Properties of Epitaxial Ge on GaAs

4.1 Photoluminescence of Ge on GaAs

Table 4.1 summarizes the growth conditions for samples analyzed in this chapter. All samples were grown on epi-ready (100) GaAs substrates without offcut. For sample 1 and 2, the growth sequence was as follows: (a) a 0.5 μm GaAs homoepitaxial layer was grown at 650 $^{\circ}\text{C}$, (b) in the absence of an AsH_3 overpressure the GaAs was annealed at 600 $^{\circ}\text{C}$ for 10 min to enable a Ga-rich surface, and (c) cooled to the Ge growth temperature. In the last Chapter, we have shown that this growth procedure, which promotes a GaAs surface with high Ga-to-As ratio, resulted in high quality Ge heteroepitaxy on GaAs.

Table 4.1. Growth condition of samples analyzed in chapter 4

	III-V Compound Growth: Template/Growth Temperature	Ge Growth Temperature	Ge Layer Thickness	Surface Preparation Procedure *
Sample 1	GaAs/650 $^{\circ}\text{C}$	450 $^{\circ}\text{C}$	600 nm	A
Sample 2	GaAs/650 $^{\circ}\text{C}$	350 $^{\circ}\text{C}$	130 nm	A
Sample 3	GaAs/650 $^{\circ}\text{C}$	450 $^{\circ}\text{C}$	300 nm	B
Sample 4	AlAs/650 $^{\circ}\text{C}$	450 $^{\circ}\text{C}$	140 nm	A

* A: 10 min annealing without AsH_3 overpressure at 600 $^{\circ}\text{C}$, cooling to Ge growth temperature without AsH_3 overpressure. B: No annealing at 600 $^{\circ}\text{C}$, cooling to Ge growth temperature with AsH_3 overpressure.

Figure 4.1 shows a cross-sectional TEM (XTEM) image of sample 1. The Ge film for this structure was grown at 450 $^{\circ}\text{C}$ and the Ge film thickness was 600 nm. AFM measurements (see Figure 4.2) showed a very low RMS roughness of 0.123 nm on a 10 $\mu\text{m} \times 10 \mu\text{m}$ scale. For sample 2 the Ge growth temperature was 350 $^{\circ}\text{C}$ and the Ge film thickness was 130 nm and the RMS surface roughness was 0.145 nm. No structural

growth defects were observed by TEM or AFM for either sample, confirming high quality Ge on GaAs heteroepitaxy. For investigation of the effects of annealing on the Ge thin films, a piece of each sample was subsequently annealed at 700 °C for 10 min in nitrogen ambient.

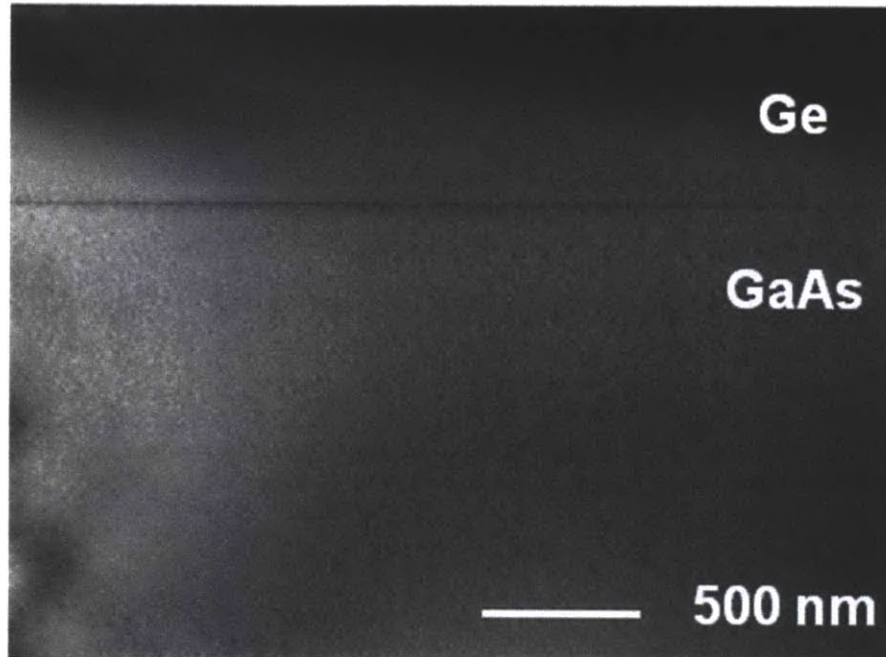


Figure 4.1. XTEM image of sample 1 showing defect free Ge layer grown on GaAs

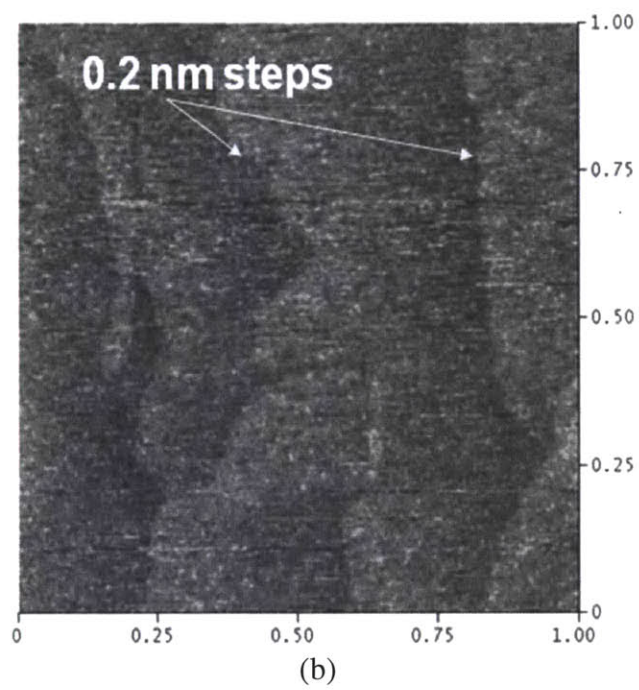
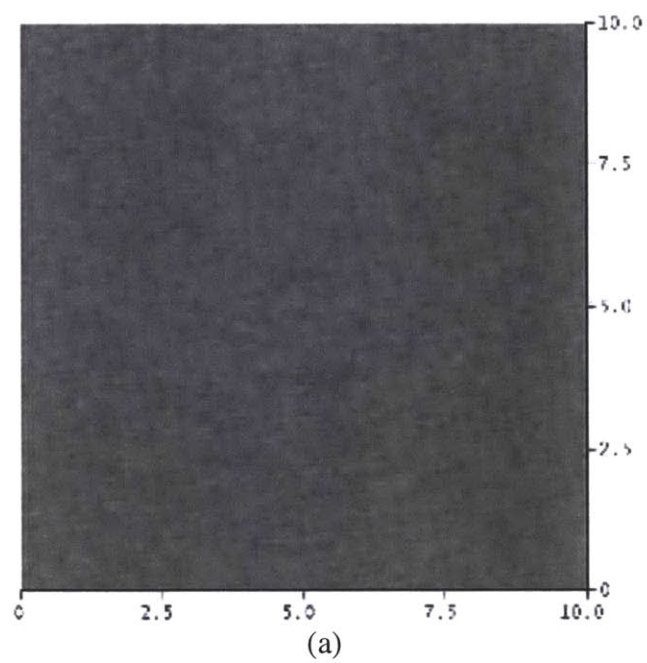


Figure 4.2. AFM image of sample 1 showing 0.2 nm tall atomic steps on the surface. a) scale: 10 $\mu\text{m} \times 10 \mu\text{m}$, RMS roughness: 0.123 nm, b) scale: 1 $\mu\text{m} \times 1 \mu\text{m}$, RMS roughness: 0.114 nm

Figure 4.3 shows the PL spectra of two samples of Ge thin films grown on epi-ready (100) GaAs substrates. PL analysis was conducted on the as-grown and annealed samples at 20 K and 300 K. Similarly to Si, at cryogenic temperatures the luminescence signal from the Ge is dominated by radiative recombination from the indirect band gap⁶¹⁻⁶⁵. At room temperature, however, the luminescence properties of Ge differ from that of Si. Due to the small energy difference between the direct and indirect band gap transitions in Ge, the widening of the electron distribution leads to a finite population of electrons into the higher energy Γ valley, enabling luminescence from Ge direct band gap. A number of reports refer to an enhancement of luminescence from Ge direct band gap with increasing temperature⁶⁶⁻⁶⁸, including Sun's recent work on PL on Ge-on-Si¹⁶. In addition, Sun's work also showed that n-type doping can be used to significantly enhance luminescence from Ge direct band gap¹⁶. N-type doping raises the Fermi-level and increases the probability for electrons occupying states in the direct Γ valley, enabling efficient band-to-band optical transitions at room temperature.

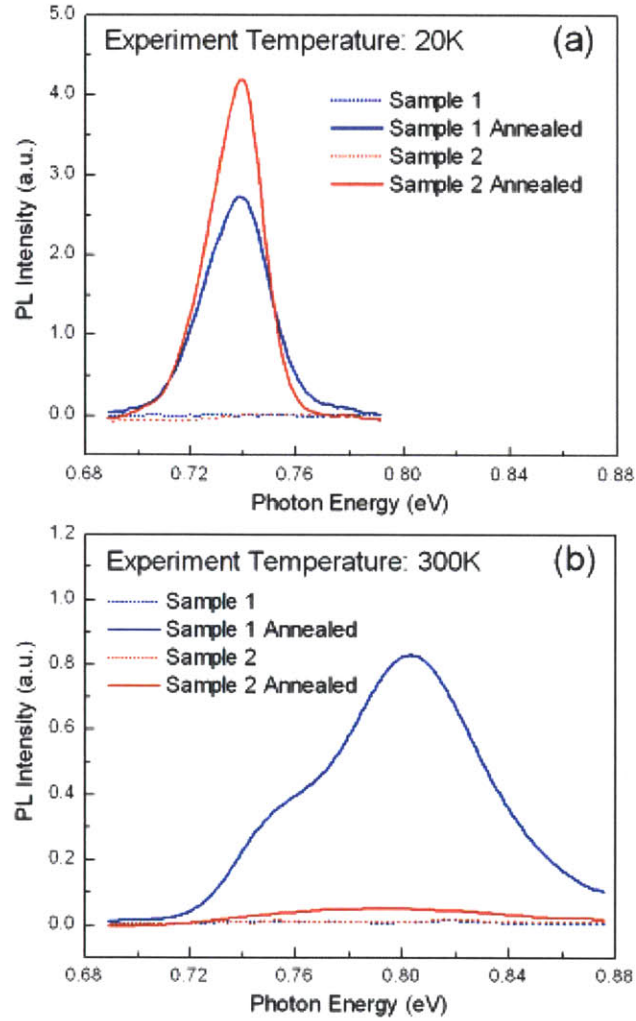


Figure 4.3. Photoluminescence (PL) spectra at 20K (a) and 300K (b) of sample 1 and sample 2 before and after annealing at 700 °C for 10 min in a nitrogen ambient. Both samples only exhibit luminescence after annealing.

As shown in Figure 4.3, both samples exhibit luminescence from the Ge indirect band gap at 20K and the Ge direct band gap at 300K. However, luminescence was only observed after annealing the samples at 700 °C for 10 min. This indicates that the Ge layer, which was grown at low temperature, had a high level of defects which were undetectable via TEM. It is likely that vacancies and other point defects in the as-grown Ge film facilitate non-radiative recombination of electron-hole pairs. The annealing at

elevated temperature improves the crystalline quality of the Ge and enhances radiative recombination.

Both samples, when annealed, show strong luminescence at 20 K. The peaks at 0.739 eV correspond well to the transverse acoustic (TA) phonon-assisted indirect $L_c-\Gamma_v$ transition in Ge.⁷⁰ More interestingly, we can see that although sample 1 exhibits weaker luminescence from the Ge indirect band gap at low temperature, at 300K its luminescence from Ge direct band gap is much stronger than sample 2. The peak at 0.803 eV corresponds well with the Ge direct $\Gamma_c-\Gamma_v$ transition. To our knowledge, the observation of the peak at 0.803 eV is the first observation of room temperature luminescence from the direct band gap of epitaxial Ge thin films grown on III-V compound templates. We should note that the peak at ~0.75 eV is not consistent with emission from the indirect gap transition since radiative recombination through the indirect band gap transition in Ge has a long recombination lifetime and is normally only observed at cryogenic temperatures.⁶⁵ Liu *et al.* reported the observation of the peak at 0.75 eV as well and suggested it was not due to a band-to-band transition⁷¹. At this time, the origin of this peak is unclear.

As we observed and others have reported, it is clear that temperature, higher excitation intensity, and n-type doping enhance luminescence from the Ge direct band gap. Since the sample 1 and 2 were tested at same temperature and same excitation intensity, the result suggested that they have different doping profiles even though neither film was intentionally doped. It is well-known that Ga and As dope Ge films p-type and n-type

respectively, and given the in situ processing we expected some memory effects of the III-V compound epitaxy to affect the electrical properties of the Ge thin films. To better understand the doping properties of the two samples we analyzed both films with SRA and SIMS.

4.2 Spreading resistance analysis of Ge on GaAs

Figure 4.4 shows the SRA depth profiles of sample 1 and 2 pre and post anneal. The active carrier concentration and type in the Ge thin film are plotted against depth from the surface. The data was calibrated against measurements on bulk Ge substrates with known resistivity and bulk Ge carrier mobilities have been assumed to calculate the carrier concentration. The accuracy of carrier concentration is within $\pm 20\%$, and sampling volume correction adds an additional uncertainty of $\pm 15\%$. The error bars in Figure 4.4 accounts for both of these uncertainties. The accuracy of depth scale is within $\pm 3\%$.

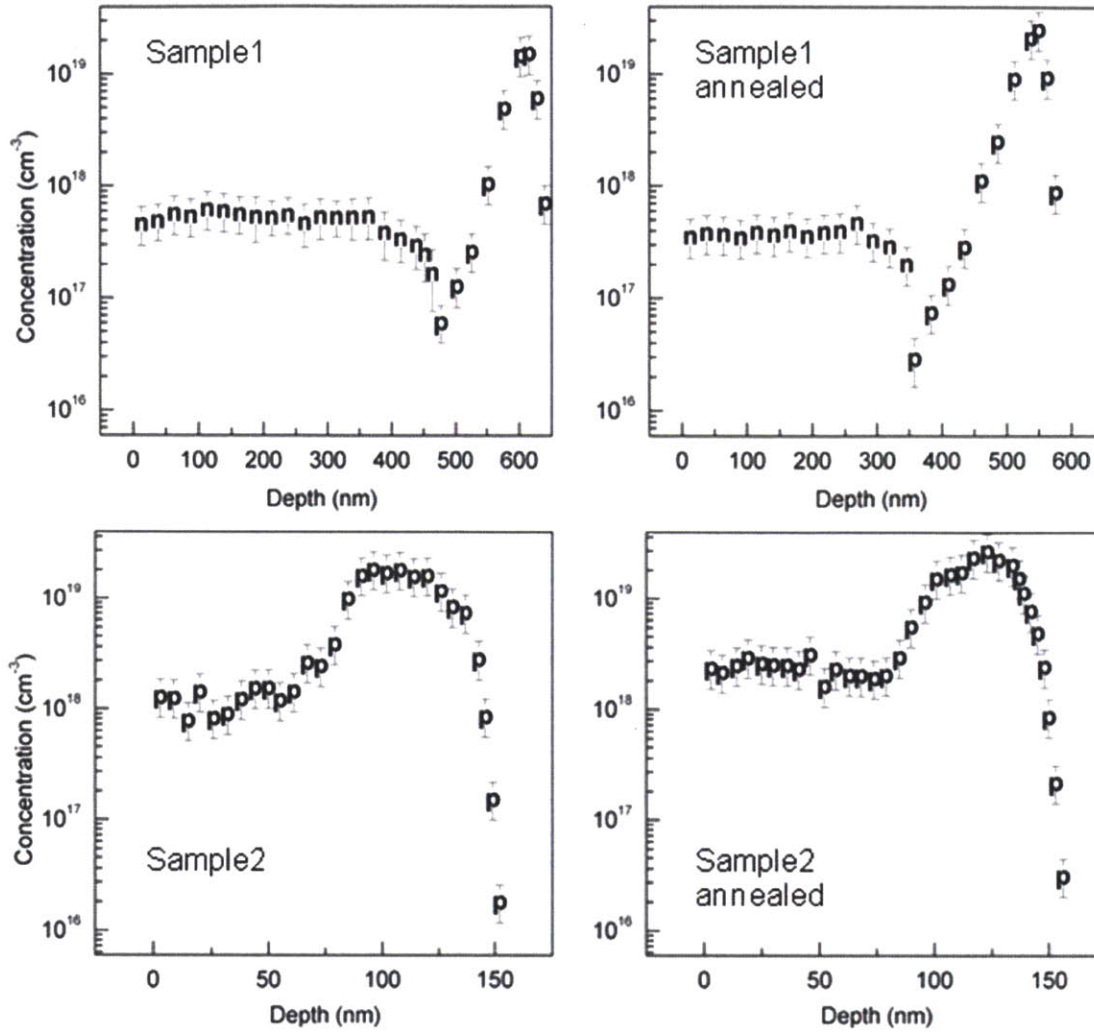


Figure 4.4. Spreading resistance analysis (SRA) results showing active carrier concentration and type for samples 1 and 2 before and after 10 min anneal at 700 °C under nitrogen ambient. Annealing does not change the doping characteristics for either sample.

It can be seen that annealing does not change the SRA characteristics for either sample 1 or sample 2. Sample 1 contains a p-n junction as it is n-type closer to the surface and p-type closer to GaAs interface, while sample 2 is p-type throughout the layer.. The depletion region for the sample with a p-n junction shows the characteristic sharp downward spike between the n and the p region in the SRA profile

The SRA data supports the PL spectra observations as the n-type Ge in sample 1 would enhance its direct band gap luminescence at 300 K, while the heavy p-type doping in sample 2 would reduce the probability of electrons populating the Γ valley, thus inhibiting direct band gap luminescence. Our results reaffirm the argument that n-type doping enhances luminescence from the Ge direct band gap.

Additionally the SRA results suggest that although annealing does not change the overall doping character of the Ge film, it does slightly enhance the p-doping levels in both samples and the change in dopant distribution is also demonstrated by the changing of the pn junction position for sample 1. We will discuss these observations in the following section.

4.3 SIMS of Ge on GaAs

We conducted SIMS analysis on the samples discussed in the previous section to determine the origin of the doping profiles measured by SRA. The SIMS profiles scanned for common Ge n-type dopants (As, P, and Sb) and p-type dopants (B, Ga, In, and Al). Care had to be taken to ensure that As and Ga species, which have atom masses very close to Ge isotopes, did not interfere with the measurements. To reduce mass interference and obtain the highest accuracy SIMS, Ga and As concentrations were scanned for separately. P, Sb, and As were scanned for with a Cs^+ ion beam, together with impurities such as C, O, N, and Si. B, Al, Ga, and In, were scanned for with an O_2^+ ion beam. With these scan conditions, the detection limits for Ga and As are $1\text{E}17$ atoms/ cm^3 and the accuracy of concentration and depth are 30% and 10%, respectively.

In general, we found concentrations of impurities other than Ga and As were below SIMS detection limit and well below the levels needed to affect the doping characteristic of the Ge film. Figure 4.5 shows the SIMS profiles of Ga and As in the Ge layer for the two unannealed samples. For easy comparison, zero in scan depth is set to the Ge/GaAs interface with the Ge layer left of the interface. Negative 600 nm on the x-axis corresponds to the Ge surface for sample 1 and negative 120 nm corresponds to the surface of sample 2. SIMS artifacts caused by the surface (~10 nm of scan length) were removed from scans.

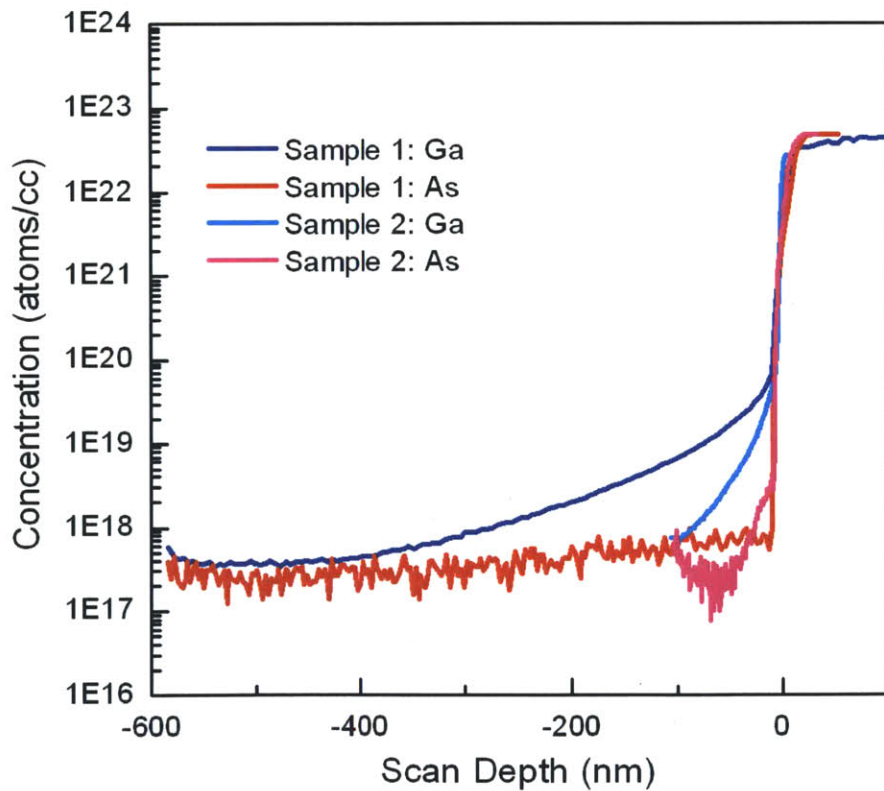


Figure 4.5. Secondary ion mass spectroscopy (SIMS) of Ga and As in the Ge epitaxial layer for sample 1 and 2

For both samples the SIMS shows the initial layer of Ge epitaxy, where SRA results in Figure 4.4 indicate highly p-type character, the Ga level is more than an order of magnitude higher than the As level. Away from the Ge/GaAs interface (i.e., as the Ge growth continued) the Ga level decreases super-exponentially. For sample 2 the high Ga level persists and is considerably higher than the As level throughout the film thickness, doping the entire Ge film p-type (the Ge film is much thinner for sample 2). For sample 1, the Ga level decreases to nearly the same level as the As concentration after 400 nm of Ge growth thickness, but remains higher than As concentration throughout. The excess presence of Ga in the SIMS seems contrary to the fact that the SRA (Figure 4.4a) shows sample 1 is n-type near the surface, and that there is a p-n junction approximately 200 nm from the Ge/GaAs interface. There are two possible cause of this inconsistency. First, although SRA and SIMS analysis when used together provide powerful insights on impurity and doping character of epitaxial thin films, the absolute values of concentrations determined by SRA and SIMS from the same sample are not directly comparable, due to the different calibration methods used in each analysis. In this case, we suspect there is a slight under estimation of As concentration in the SIMS analysis. Secondly, SRA measures the electrically active carrier concentration, while SIMS measures concentration of an impurity element regardless whether is electrically active or not. We believe that the Ga concentration incorporated in the Ge layer in the as-grown sample is less than 100% electrically active, due to the relatively low growth temperature.

A comparison of Ga and As SIMS profiles in sample 1 before and after annealing (Figure 4.6) also supports this argument. We can see that after annealing at 700 °C for 10 min,

the Ga and As concentration had nearly no change throughout the Ge film, except for the first 40 nm of Ge film close to the Ge/GaAs interface. Nevertheless, as point out in the last section, a comparison of Figure 4.4a and Figure 4.4b suggests that after annealing, the n-type doping of the n-type part of the Ge film close to the surface decreased slightly, while the p-type doping of the p-type part of the Ge film close to the Ge/GaAs interface increased slightly, and the p-type region became slightly wider, moving the p-n junction closer to the surface. Here it's worth pointing out that direct comparison of absolute value of concentration of two SRA scans is valid and even the slightest change suggests actual difference in the samples, because all the samples were scanned with exact same condition and data extraction was done with exact same calibration. The same holds for comparing SIMS profiles. So the above analysis indicates after annealing the Ge thin film in Sample 1 overall becomes more p-type throughout the film, while there's no change in total concentration of Ga. The only logically explanation is that not the entire population of the incorporated Ga was electrically active in the as-grown sample, and the annealing process activates more Ga to be electrically active.

Through the above analysis, we clearly established that Ga and As incorporation and compensation determines the overall doping profile of the Ge layers grown on GaAs, which affects the luminescence properties of the Ge film.

4.4 Ga incorporation in Ge epitaxial layer

The Ga and As levels in sample 1 (annealed and unannealed) shown in Figure 4.6 indicate that Ga and As do not incorporate into Ge film through solid-state diffusion since the annealing process (10 min at 700 °C), which is much higher thermal budget than the

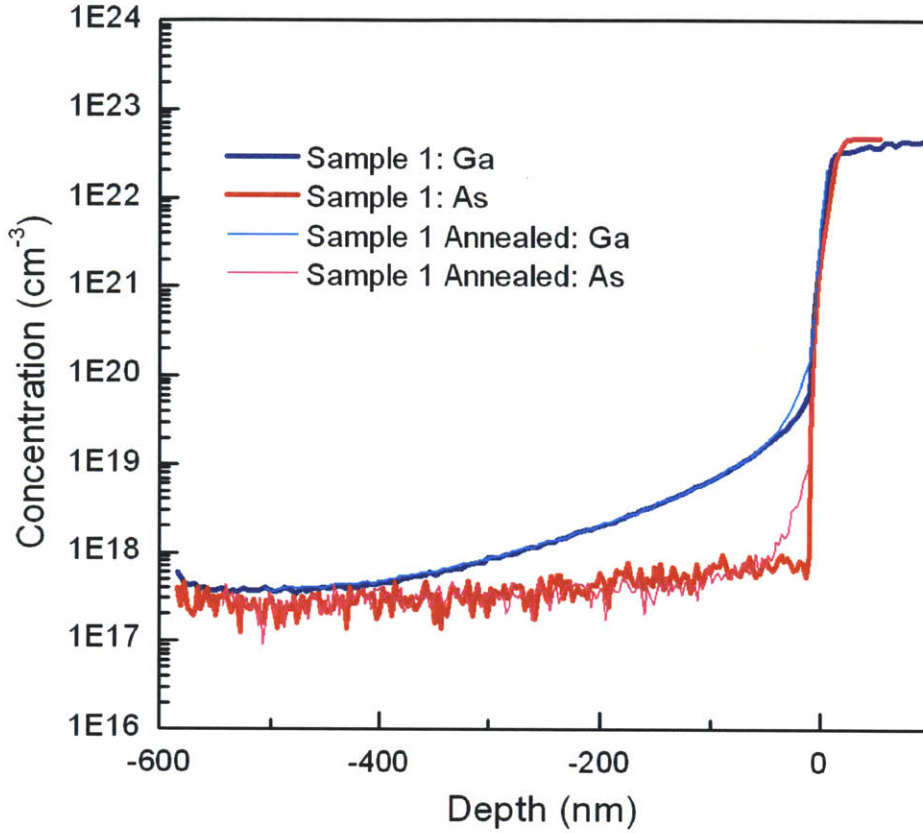


Figure 4.6. Comparison of SIMS profile of Ga and As concentration in the Ge epitaxial layer of sample 1 before and after 10 minutes anneal at 700 °C under nitrogen ambient. After the anneal, with the exception of the first 40 nm close to the Ge/GaAs interface, the Ga and As concentrations remains unchanged throughout the Ge layer.

initial growth process (10 min at 450 °C), only changed the Ga and As profiles marginally and its effects were limited to the first 50 nm of Ge film thickness. As illustrated in Figure 4.7, the change in the Ga concentration profile in the first 50 nm of the Ge film after annealing can be modeled well with a typical error function diffusion profile, assuming the GaAs substrate act as a saturated Ga source, according to the following equation.

$$C(x,t) = C_{ss,Ga} \operatorname{erfc}\left(\frac{x}{2\sqrt{D_{Ga}t}}\right) \quad \text{Eq. 4.1}$$

Here $C(x,t)$ is the concentration of Ga in the Ge layer at position x , after annealing time t ; x is the distance into the Ge layer from the Ge/GaAs interface; $C_{ss,Ga}$ is the solid solubility of Ga in Ge, which is about $2.7 \times 10^{20} \text{ cm}^{-3}$,⁷² and D_{Ga} is the diffusivity of Ga in Ge at the annealing temperature, which is approximately $3.31 \times 10^{-15} \text{ cm}^2/\text{s}$.^{73,74}

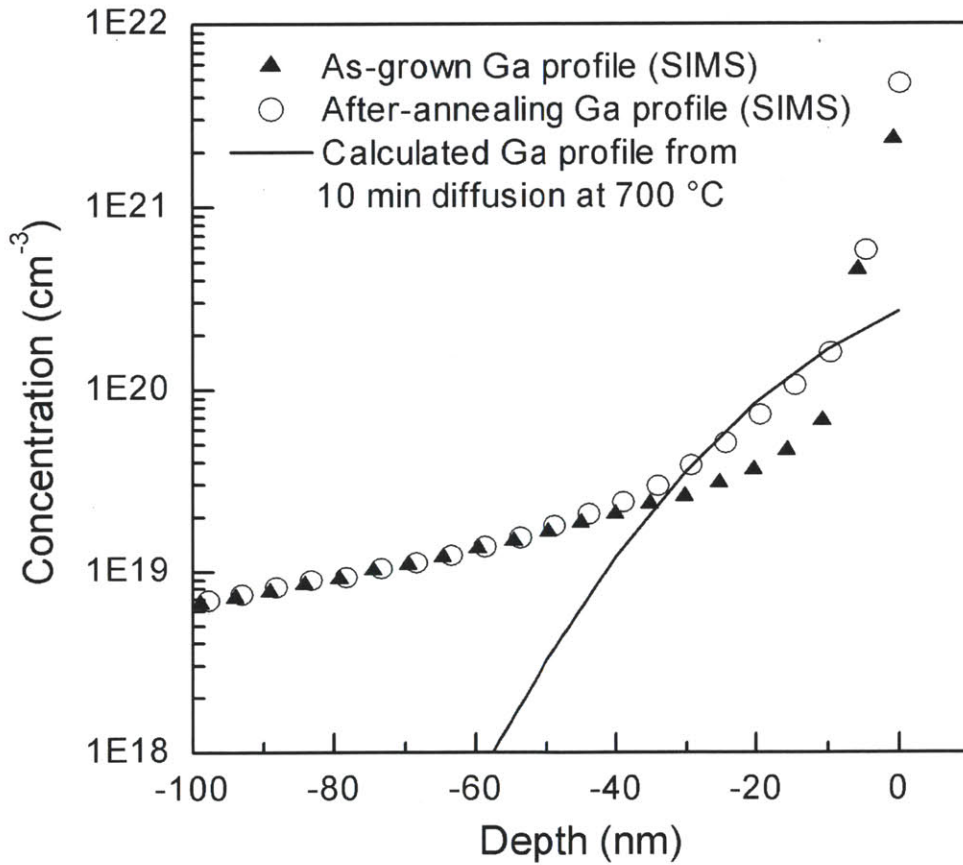


Figure 4.7. Replot of SIMS profile of sample 1 Ga profile in the Ge layer close to the Ge/GaAs interface before ($\blacktriangle$) and after ($\circ$) 10 min anneal at 700 °C, and comparison to calculated Ga level (solid line) assuming error function diffusion profile, and saturation of Ga in Ge layer at Ge/GaAs interface. It is apparent that thermally activated diffusion cannot account for the high Ga levels that are present throughout the as-grown Ge layer.

Our analysis indicates that solid-state diffusion accounts for the Ga concentration increase in the first 50 nm of Ge film close to the Ge/GaAs interface, but is not accountable for the very high levels of Ga in the entire Ge layer present after a relatively low temperature epitaxy sequence. This Ga incorporation during low temperature epitaxy is due to another incorporation mechanism.

Figure 4.8 compares the Ga and As SIMS profiles in the Ge film for sample 1 to another sample, sample 3.

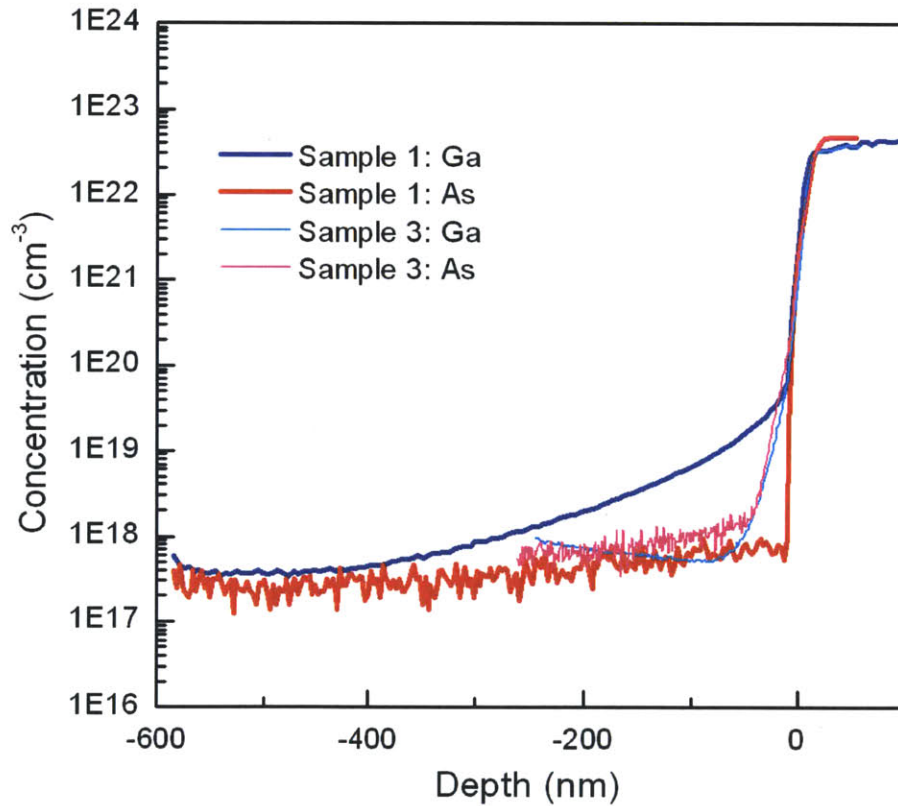


Figure 4.8. Comparison of SIMS profile of Ga and As in the Ge epitaxial layer of sample 1 and sample 3.

For sample 3, the Ge was grown at the same temperature as sample 1, but the initial GaAs surface had a low Ga-to-As ratio (i.e., the surface was As rich). For sample 3, after GaAs homoepitaxy at 650 °C, the sample was cooled under an AsH₃ flow to 450 °C prior to Ge growth. The AsH₃ overpressure was in place until 10 seconds prior to introducing GeH₄ (initiation of Ge growth) into the growth chamber. This procedure preserves the typical GaAs surface structure obtained during GaAs MOCVD, which is known to be a c(4×4) or d(4×4) surface reconstruction with the presence of excess As (low Ga-to-As ratio) on the surface.⁴⁶ Our previous work showed that Ge thin films initiated on a GaAs surface with a low Ga-to-As ratio led to incomplete Ge layer coverage at initial stage and resulted in defective Ge layer, so we did not use this approach initially in this work. However, for the purposes understanding the doping mechanisms during Ge epitaxy, we examined this sample. Figure 4.8 shows that in sample 3, the Ga concentrations are much lower than in sample 1 and in the part close to the Ge/GaAs interface it is even lower than the As concentrations levels. SRA analysis of sample 3 (not shown here) also confirms that Ge layer in Sample 3 is n-type throughout its thickness. After annealing the sample for 10-min at 700 °C, PL spectra of sample 3 shows luminescence from the indirect band gap at 20K and the direct band gap at 300K, which is consistent with the luminescence properties of other films with a n-type character.

Based upon the data, we hypothesize that the GaAs surface Ga-to-As ratio plays an important role in the ultimate Ga and As profiles in the subsequent Ge epitaxial film. When As atoms are driven off the surface, the stable surface reconstruction for a (100) GaAs surface with high Ga-to-As ratio is known to be (4x2) reconstruction with $\frac{3}{4}$

monolayer of Ga dimerized along the [110] direction^{46,75}. The Ga atoms on the surface serve as available binding sites for Ge adatoms to form Ga-Ge dimers, which have a lower formation energy than a As-Ge dimers⁷⁶. The more favorable energetics for a Ga-rich GaAs surface facilitates the initiation of Ge epitaxy. However, we postulate that in conjunction with the Ga-Ge dimer formation, the Ga atoms exchange with the Ge atom and segregate on the growth surface to lower surface energy. Thus, during the epitaxy the surface segregated Ga atoms would be gradually incorporated into the Ge layer. If we integrate the Ga concentration in sample 1 over the entire Ge film thickness we calculate a sheet Ga density (dose) of $4.15 \times 10^{14} \text{ cm}^{-2}$, which is equal to 66.4% of the sheet density of one monolayer of Ga on a (100) GaAs surface. This suggests that a majority of the Ga atoms on the (4x2) reconstructed surface take part in the exchange process during the ongoing Ge deposition. Similar surface segregation phenomena is well known for Si layers deposited on SiGe surfaces, where the Ge from the SiGe layer would segregate to the growing Si surface and result in Ge incorporation in the Si layer with a similar decaying concentration profile to what we observed here for Ga in depositing Ge film⁷⁷⁻⁸⁰. This type of process has been modeled with a two-state exchange kinetics model⁸⁰, modified from a dopant segregation exchange model^{81,82}.

From our analysis, we believe that a large percentage of Ga atoms on the Ga-rich GaAs surface are incorporated into the Ge film but a portion are electrically inactive due to the low growth temperatures, between 350 °C and 450 °C. Upon annealing at higher temperature, 700 °C, more Ga atoms become electrically active, resulting in the increase

of active p-type carrier concentration observed with SRA, and increasing the p-type character of the Ge epitaxial layer.

We observed similar behavior for Ge epitaxial layers grown on AlAs layers where Al incorporated into the Ge film. Figure 4.9 shows the Al and As concentration profiles in a Ge epitaxial layer deposited on an AlAs surface (sample 4). A 120 nm AlAs layer was grown after GaAs homoepitaxy at 650 °C. Following the same high III-to-V ratio surface preparation procedure for samples 1 and 2, the surface was annealed at 600 °C in the absence of AsH₃ for 10mins, before cooling to the sample to 450 °C for Ge epitaxy. It can be seen in Figure 4.9 that the Al concentration levels in the Ge epitaxial layer close to the Ge/AlAs interface are high and exceed the As concentration level and the Al concentration decreases but persists for the rest of the film. After the sample was annealed at 700 °C for 10 min, PL spectra of sample 4 revealed luminescence from Ge indirect band gap at 20K but not from Ge direct band gap at 300K, indicating that it is of p-type character. These results reinforce our theory that a surface preparation technique that promotes a group III rich surface leads to excess group III (p-type) elements being incorporated into Ge, which in turn suppresses luminescence from the direct band gap transition.

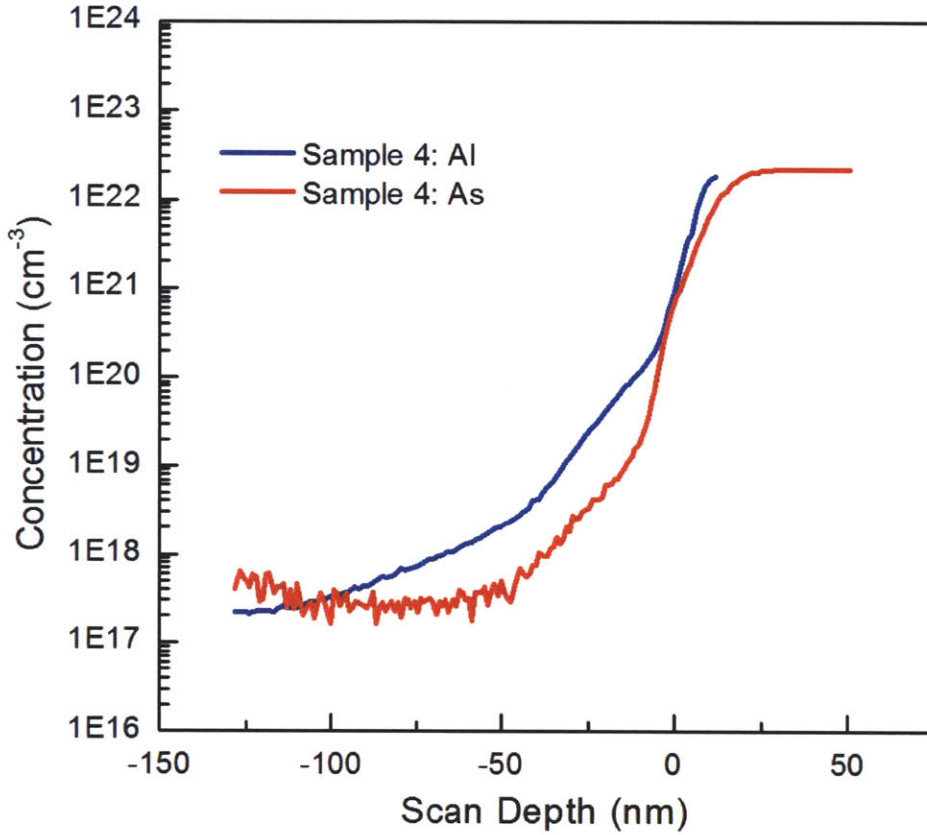


Figure 4.9. SIMS profile of sample 4 showing Al and As concentrations in a Ge epitaxial layer grown on an AlAs surface.

It is interesting to note that the same surface exchange mechanism is not prevalent for a As-rich surface. An As-rich GaAs surface, sample 3 for example, would have either a $c(4\times4)$ or $d(4\times4)$ reconstruction with multiple layers of As dimers on the surface which would also be more than the amount of Ga dimers on a Ga-rich GaAs surface. Looking at the As concentration profiles for sample 3 in Figure 4.8, the same super-exponentially decaying profile does not exist for the As concentration levels close to Ge/GaAs. We believe that since Ge-As dimer formation costs more energy and As can readily leave the

surface through gas phase transport in the form of As_2 , As_4 , or AsH_3 , As is less likely to be trapped in the Ge during film growth. In fact regardless of the Ge initiation conditions, the As concentration levels plateau at nearly the same value, mid 10^{17} cm^{-3} , regardless of the underlying III-V layer and film thickness. This suggests that the As concentration in the Ge film far from the interface is due to background levels of volatile As species within the reactor ambient. We estimated a background As_4 over pressure of 1×10^{-6} - 1×10^{-5} Torr (1.3×10^{-4} - 1.3×10^{-3} Pa). Incorporation of As through gas phase transport can pose a challenge when attempting to control the background As level in the epitaxial Ge thin film grown on III-V compound templates. For some applications the background As can be provide a benefit, as mentioned throughout this paper n-type doping can enhance the optical properties of Ge. For cases where high background As concentration levels are a hindrance, procedures would have to be developed to reduce the partial pressures of As_2 and As_4 . For example, a system configuration with connected group IV and III/V growth chambers would provide excellent control of the background impurity. However it would require costly redesign and modification of the entire system. Intentional doping can also be applied to compensate and mask out the unintentional doping, although additional effects of high compensation level are not entirely clear.

It is interesting to note that our results point to a trade-off between surface morphology and ‘auto-doping’. To achieve planar Ge films at the lowest possible thickness, Ga-rich surfaces are desired. Such surfaces avoid the ‘pitting problem’ which is due to a high degree of surface diffusion and the tendency for Ge to bond to itself when the surface is As-rich. However, Ga-rich surfaces result in much Ga exchange, resulting in high Ga

concentrations in the Ge film, and apparently a large concentration of it is not electrically active. This trade-off appears fundamental in that excellent surface morphology is achieved when there is a larger affinity for like-species (Ga and Ge) but a concomitant increase in exchange is unavoidable without further process modification.

4.5 Summary

In this chapter, we investigated the correlation between growth conditions, auto doping and optical properties of Ge thin films grown on GaAs and AlAs compounds. The results from this investigation helped us develop further understanding on the surface process of Ge epitaxy on III-V surface. It's likely that Ge epitaxy initiates by bonding to and exchange with the surface group III atoms, which explains our observation in last chapter that Ga-rich surface facilitates fast initiation and high quality Ge epitaxy. As a result of this mechanism, elevated levels of Group III element, Ga and Al, concentrations in the Ge layer occur through surface segregation during Ge epitaxy and it strongly depends on the III-to-V ratio of initial III-V surface. Group V, As, concentrations are mainly dependent on gas phase transport of volatile As_2 and As_4 in the growth ambient.

Incorporation of both group III and group V elements generate highly doped and compensated Ge epitaxial thin films, and in addition to low growth temperature, possibly increase the point defect density in the Ge layer. Luminescence studies of Ge/III-V thin films stacks showed that the low growth temperatures for the Ge epitaxy inhibited the activation of the doping species and resulted in point defect concentrations that suppressed radiative recombination. A post growth annealing sequence was required to observe luminescence from the Ge indirect band gap at 20K and the Ge direct band gap

transition at 300K. Additionally, the Ge film had to have an n-type doping character to observe luminescence from Ge direct band gap at room temperature. In this case, high level of unintentional As doping prove to be beneficial. To our knowledge, this is the first observation of room temperature luminescence of the Ge direct band gap transition from Ge epitaxial thin films grown on III-V compound substrates.

Chapter 5 Highly Tensile Strained Ge Grown on $\text{In}_x\text{Ga}_{1-x}\text{As}$

5.1 Previous methods of fabricating strained Ge and their limitations.

Two methods have been investigated by different research groups to produce tensile-strained Ge. The first method utilizes the thermal mismatch between Ge and Si to induce tensile strain in Ge thin films grown on Si substrates by cooling it from high annealing temperatures. Through this method, enhanced near-IR absorption was demonstrated for Ge detector applications.^{13,83} However, the amount of tensile strain applied by the thermal stress was limited to 0.2%, far below the strain level needed for the direct /indirect band gap crossover. The second method proposed to induce tensile strain in Ge was by depositing Ge on relaxed $\text{Ge}_x\text{Sn}_{1-x}$. Unfortunately, the Ge-Sn alloy system is inherently unstable and it is difficult to achieve high quality GeSn epitaxy^{84,85}. Although over the years some progress has been made in Si-Ge-Sn technology, this method has not succeeded in achieving direct-band-gap Ge because the strain induced through Ge epitaxy on Ge-Sn alloys has been limited to about 0.5%^{86,87}. Takeuchi *et al.* reported the highest achieved tensile strain level of 0.68% in Ge grown on GeSn buffer⁸⁸. However the quality of this tensile-strained Ge film is limited by the quality of GeSn buffer.

5.2 $\text{In}_x\text{Ga}_{1-x}\text{As}$ compositionally graded buffers as a virtual substrate for tensile-strained Ge growth.

To induce tensile strain via epitaxy, Ge has to be grown on a stable epitaxial template that possesses a lattice constant larger than that of Ge (5.658 Å). The $\text{In}_x\text{Ga}_{1-x}\text{As}$ alloy system has lattice constants that vary from the GaAs lattice constant (5.653 Å) to the InAs lattice constant (6.058 Å). High quality $\text{In}_x\text{Ga}_{1-x}\text{As}$ compositionally graded buffers (GB) have

been grown by metal-organic chemical vapor deposition (MOCVD)⁸⁹ and molecular beam epitaxy (MBE) for integration of InP-based devices on GaAs^{90 91}. These graded buffers are thermodynamically stable and the threading dislocation density (TDD) can be on the order of 10^6 cm^{-2} , providing a platform suitable for electronic and optoelectronic device integration. Therefore, a relaxed $\text{In}_x\text{Ga}_{1-x}\text{As}$ compositionally graded buffer grown on GaAs substrate can work as a strain-inducing template for tensile-strained Ge thin film growth. The amount of lattice mismatch between $\text{In}_x\text{Ga}_{1-x}\text{As}$ and Ge can be controlled by varying the indium composition. Other alloys such as $\text{Al}_x\text{In}_{1-x}\text{As}$ can be grown lattice-matched to the graded buffer for device fabrication purposes.

5.3 Growth procedures

Extending our results of Ge growth on GaAs, we studied the growth of tensile-strained Ge on $\text{In}_x\text{Ga}_{1-x}\text{As}$. To grow Ge/ $\text{In}_x\text{Ga}_{1-x}\text{As}$ heterostructures, we started with (100) GaAs substrates and grew a GaAs homoepitaxial layer followed by the $\text{In}_x\text{Ga}_{1-x}\text{As}$ compositionally graded buffer at 700 °C. Further details of growth methods for the $\text{In}_x\text{Ga}_{1-x}\text{As}$ compositionally graded buffers can be found elsewhere.^{89,91} The compositionally graded buffers were capped with relaxed uniform composition $\text{In}_x\text{Ga}_{1-x}\text{As}$ layers with $\text{TDD} < 5 \times 10^6 \text{ cm}^{-2}$, which serve as the strain-inducing templates for the tensile-strained Ge growth. The final In composition in these uniform composition layers ranged from 5% to 25%. The GBs were then capped with a 20Å, strained GaAs layer. This strained GaAs layer reduces the critical thickness of the subsequent tensile-strained Ge layer. Nevertheless, it retains the studied Ge/GaAs interface for Ge initiation, although strain has been incorporated.

We then adopted the annealing method described earlier to prepare this surface for Ge layer initiation. However, after the uniform-composition-In_xGa_{1-x}As/strained-GaAs surface was annealed at 650 °C without AsH₃ overpressure, we observed defects on the surface as shown in Figure 5.1. These defects destroyed the integrity of the surface and blocked the growth of the subsequent Ge films. We suspect that evaporation enhances InGaAs/GaAs interdiffusion, and In-rich areas form and locally melt. Even the 20Å strained GaAs layer does not offer enough protection. The defects form during the melting and re-solidifying of part of the surface. After we reduced the annealing temperature to 550 °C to prepare the surface, the formation of the melting defects was eliminated.

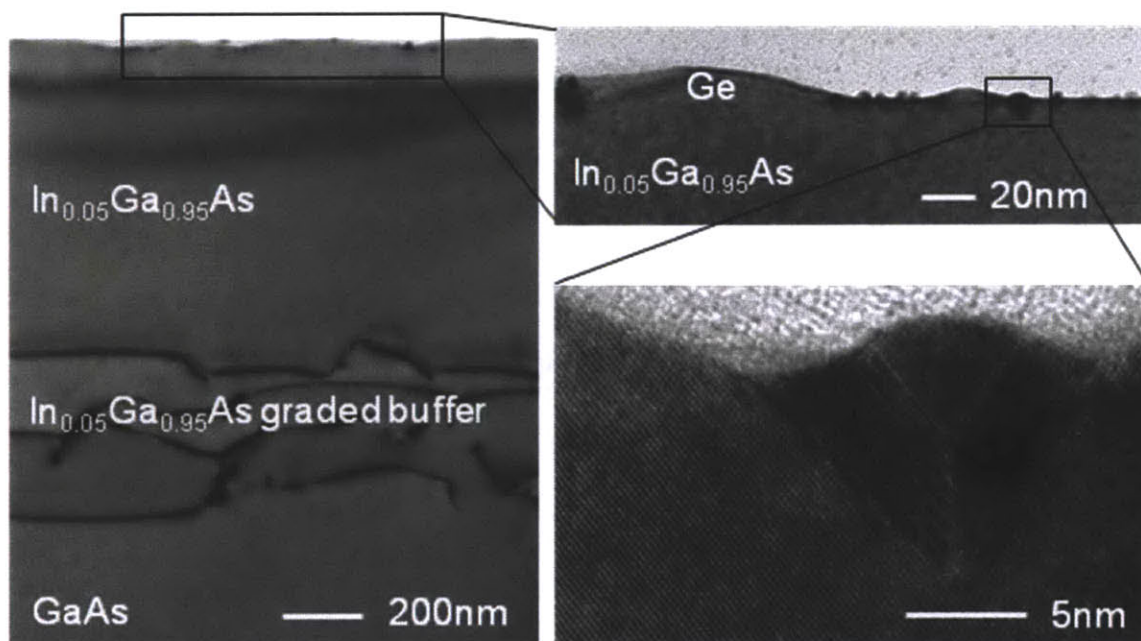


Figure 5.1. XTEM images showing Ge growth blocked by defects on In_{0.05}Ga_{0.95}As surface generated from annealing at 650 °C

With the surface preparation step modified to annealing at 550 °C, we were able to grow defect-free tensile-strained Ge thin films on In_xGa_{1-x}As graded buffers with varying In

content, as shown in Figure 5.2. All Ge films were controlled far below critical thickness to prevent strain relaxation (see Appendix 1, Appendix 2). For Ge grown on graded buffers with low In content ($x_{\text{In}} \sim 5\%$, Figure 5.2a), the film was perfectly flat and free of extended defects. At intermediate In content ($x_{\text{In}} \sim 10\%$, Figure 5.2b), the film was free of defects but was slightly undulated due to strain-driven surface diffusion. At high In content ($\sim 20\%$), the mismatch was so high that the Ge epi-layer formed quantum dots to accommodate the large amount of strain.

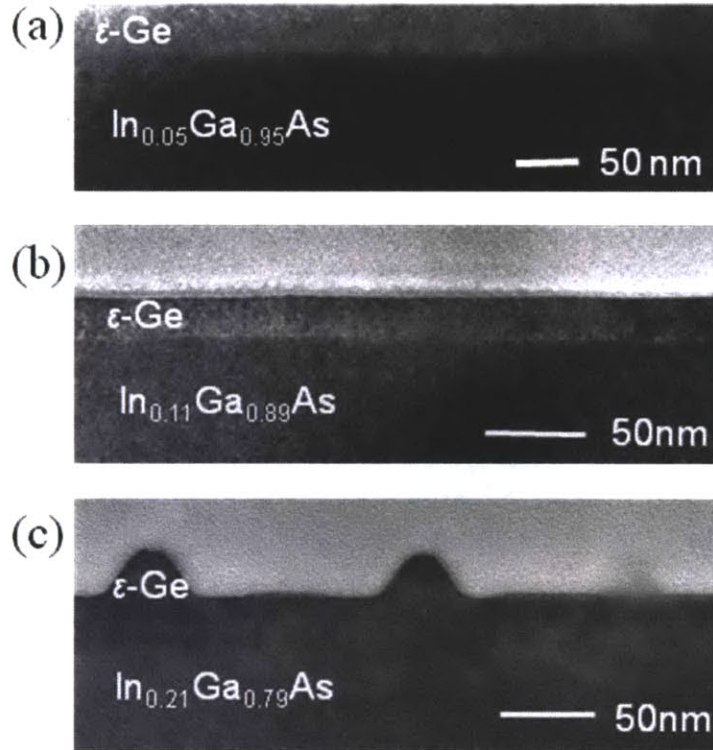


Figure 5.2. XTEM images of tensile strained Ge layers grown on $\text{In}_x\text{Ga}_{1-x}\text{As}$ with different In content: (a) 5%, (b) 11%, (c) 21%.

Interestingly, AFM images (Figure 5.3a) show that all the nucleation of Ge quantum dots occurred at the step bunches on the $\text{In}_x\text{Ga}_{1-x}\text{As}$ surface (Figure 5.3b). In general, no misfit dislocations were found at the Ge/InGaAs interface, suggesting that there was no plastic

strain introduced and the Ge structures were elastically strained to match the lattice constant of the virtual substrate. In the case of the islands, some strain relief will occur due to the ability for the island to relieve elastic strain at the surface of the island. The tensile strain in the planar films was later confirmed by experiment.

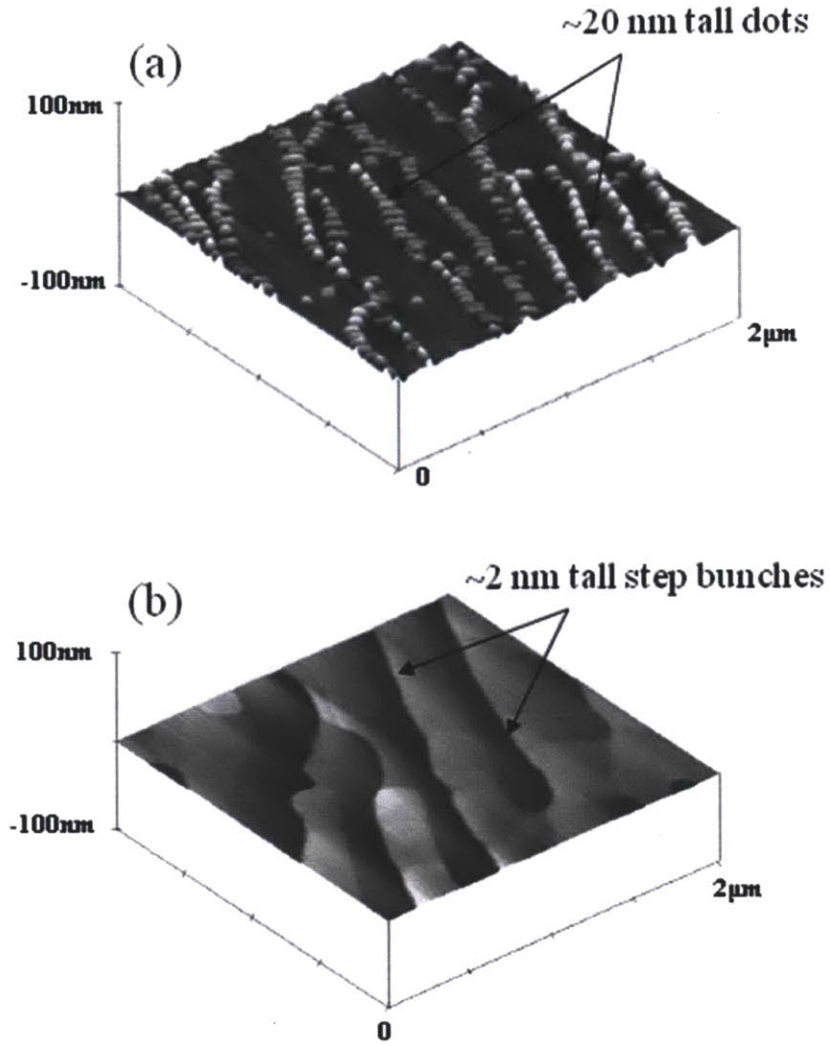


Figure 5.3. (a) AFM images of Ge grown on $\text{In}_{0.21}\text{Ga}_{0.79}\text{As}$, same sample as in Figure 5.2c. Nucleation of tensile Ge quantum dots occurs at the step bunches on $\text{In}_{0.21}\text{Ga}_{0.79}\text{As}$. (b) AFM images of an $\text{In}_{0.21}\text{Ga}_{0.79}\text{As}$ graded buffer surface revealing the step bunches.

5.4 Strain measurement by Raman spectroscopy

We grew thin tensile-strained Ge layers to stay well below the thermodynamic critical thickness, which made it difficult to measure the strain in these Ge layers via XRD. Instead, we used Raman spectroscopy to measure the strain levels in these Ge layers. Raman spectroscopy is widely used for stress measurement in silicon integrated circuits⁹², and the (100) back scattering configuration used here is very convenient for measuring strain in very thin films. For this configuration, only longitudinal optical (LO) phonons provide an active Raman mode. The relation between the shift in this Raman mode, $\Delta\omega$, and the biaxial stresses, σ_{xx} and σ_{yy} , is given by the following equation⁹²:

$$\Delta\omega = \frac{1}{2\omega_0} [(pS_{12} + q(S_{11} + S_{12}))](\sigma_{xx} + \sigma_{yy}) \quad \text{Eq. 5.1}$$

Here ω_0 is the frequency of this Raman mode without biaxial stress. S_{ij} are the elastic compliance tensor elements of Ge. p and q are phonon deformation potentials for Ge. If $\sigma_{xx}=\sigma_{yy}$, this equation can be simplified as $\Delta\omega=b\varepsilon_{\parallel}$ with $\varepsilon_{\parallel}$ as biaxial tensile strain and $b = [m - n(C_{12}/C_{11})]/\omega_0$ ⁸⁷. In Ref. 87, Fang *et al.* calculated b to be $-415 \pm 40 \text{ cm}^{-1}$ and used this method to measure tensile strain in Ge films grown on $\text{Ge}_{0.975}\text{Sn}_{0.025}$.

Figure 5.4 shows Raman spectrum data obtained from tensile-strained Ge films grown on $\text{In}_x\text{Ga}_{1-x}\text{As}$. Given b is a negative number, tensile strain causes a down-shift of the Raman mode. For the continuous tensile-strained Ge thin films grown on $\text{In}_{0.11}\text{Ga}_{0.89}\text{As}$, the measured shift from bulk Ge Raman peak, $\Delta\omega$ is $-2.4 \pm 0.1 \text{ cm}^{-1}$. The error comes from the resolution limit of the Raman setup.

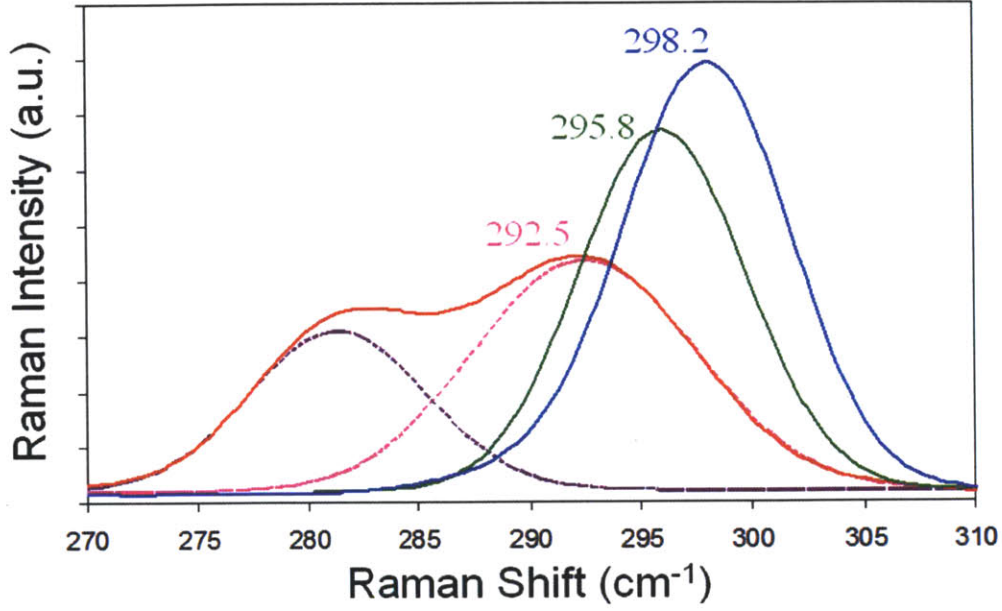


Figure 5.4. Room temperature Raman spectrum of bulk Ge (blue), Ge thin film grown on $\text{In}_{0.11}\text{Ga}_{0.89}\text{As}$ (green) and Ge QDs grown on $\text{In}_{0.21}\text{Ga}_{0.79}\text{As}$ (red). For the Ge QDs grown on $\text{In}_{0.21}\text{Ga}_{0.79}\text{As}$, the spectrum consists of both an $\text{In}_{0.21}\text{Ga}_{0.79}\text{As}$ peak (purple dotted line) and a strained Ge (pink dotted line), due to incomplete surface coverage by tensile-strain Ge QDs.

Using this value and $b = -415 \pm 40 \text{ cm}^{-1}$ from Ref. 87, we deduce the level of tensile strain in this Ge film to be $\varepsilon_{||} = 0.58\% \pm 0.08\%$. For the tensile-strained Ge QDs grown on $\text{In}_{0.21}\text{Ga}_{0.79}\text{As}$, due to incomplete QD surface coverage, the Raman spectrum is composed of the tensile-strain Ge peak and the $\text{In}_{0.21}\text{Ga}_{0.79}\text{As}$ peak. The LO phonon energy in $\text{In}_{0.21}\text{Ga}_{0.79}\text{As}$ is about 34.5 meV (linearly interpolated from GaAs and InAs⁹³), very close to the LO phonon energy of Ge, 37 meV. Therefore the two peaks are very close together and overlap with one another. After deconvolution, the $\Delta\omega$ for tensile-strained Ge QDs was found to be $-5.7 \pm 0.1 \text{ cm}^{-1}$ and the corresponding level of tensile strain is $1.37\% \pm 0.15\%$. The measured strain levels in all samples are plotted in Figure 5.5 against the effective In content in the $\text{In}_x\text{Ga}_{1-x}\text{As}$ cap layer. An “effective In content” was used to compensate for the fact that the uniform composition $\text{In}_x\text{Ga}_{1-x}\text{As}$ layers were not 100% relaxed as measured by XRD reciprocal space maps. The uncertainty in the measured

strain comes from the measurement resolution limit and the error in the value of b . Figure 5.5 shows that the actual strain levels in the Ge epi-layers are very close to the theoretical value of lattice mismatch. This confirms that essentially no strain relaxation occurred, as suggested by the XTEM results from Figure 5.2. We note here that the planar, $\sim 0.6\%$ tensile-strained Ge film is already interesting for FET applications. Such a film should support an electron and a hole mobility of $3000 \text{ cm}^2/\text{v}\cdot\text{s}$. Such a symmetric, and high, mobility is not currently available in any other single semiconductor material.

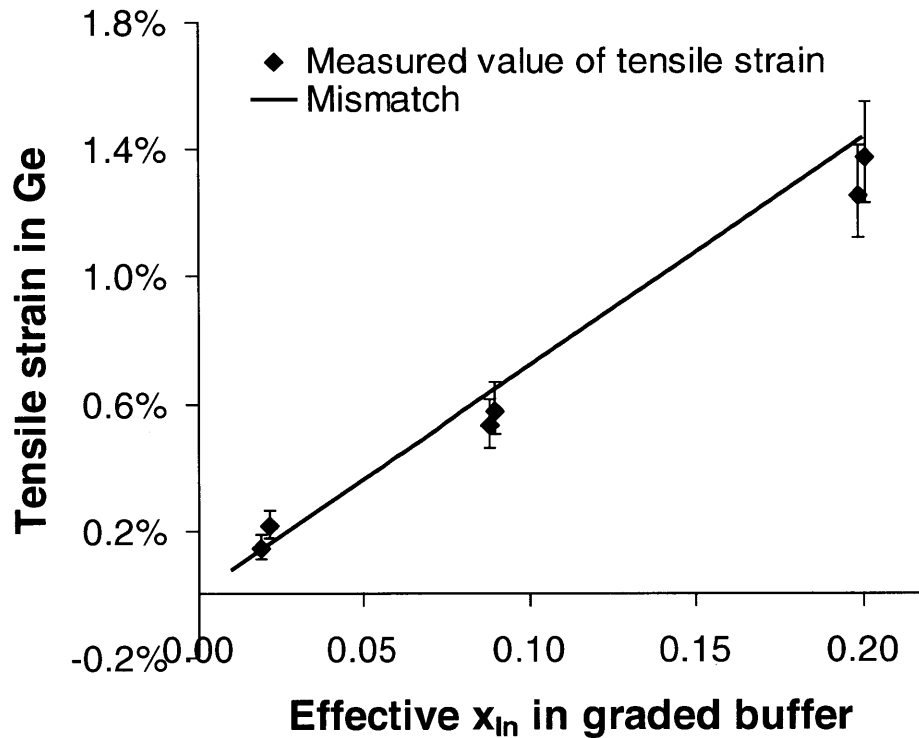


Figure 5.5. Comparison of strain levels measured by Raman spectroscopy in tensile-strained Ge layers and the theoretical lattice mismatch between Ge and $\text{In}_x\text{Ga}_{1-x}\text{As}$

Although we have successfully induced a large amount of tensile-strain, the $\text{In}_x\text{Ga}_{1-x}\text{As}$ underlayer also created a high level of unintentional doping in the tensile-strained Ge thin film, which severely hinder our ability to control the electronic properties of the Ge thin

film. Our repeated attempt to measure room temperature carrier mobility of the Ge layer was consistently affected by the uncontrolled doping profiles. In addition, the photoluminescence study of the tensile-strained Ge thin films and QD layers exhibited spectra that were dominated by the III-V layer, possibly due to the very small thickness of the Ge layer. Future development of Ge/III-V device structures will focus on control of the unintentional doping during the epitaxy process and tailor the epitaxial sequence to maximize control over unintentional doping for specific target device structures.

5.5 Summary

In this chapter we demonstrated that the same initiation mechanism for Ge on GaAs holds for Ge epitaxy on $\text{In}_x\text{Ga}_{1-x}\text{As}$. Modifying the surface preparation step from 650 °C to 550 °C to avoid local melting of the $\text{In}_x\text{Ga}_{1-x}\text{As}$ surface, we have successfully grown structurally high quality tensile-strained Ge thin films and quantum dots on $\text{In}_x\text{Ga}_{1-x}\text{As}$. Low growth temperatures between 350 °C and 500 °C suppress strain relaxation via formation of misfit dislocations. Tensile strain levels as high as 0.58% in Ge thin films and 1.37% in Ge quantum dots were achieved, far higher than previously achieved values via other methods. However our ability to control the electronic properties of tensile-strained Ge thin films requires further modification of the process steps to inhibit the high level of unintentional doping caused by surface exchange.

Chapter 6 Ge/III-V Heterostructures for Engineered Substrates

6.1 State-of-Art of engineering substrates fabrication

All engineered substrates, e.g. SOI, GeOI, and SOLES, consist of thin layer(s) of high quality semiconductor materials on top of single or multiple oxide layers. Fabrication of these engineered substrates is enabled by wafer bonding and layer exfoliation technologies. So far, SmartcutTM technology has been demonstrated as the most advanced layer exfoliation technique used in fabricating these engineered substrates. However, SmartcutTM is also an expensive process due to the requirement of hydrogen implantation.⁹⁴ It also has a limited process temperature window and encounters thermal stress issues when applied to large wafer size and materials with very different coefficients of thermal expansion (CTE).^{95,96} Thus, there exists an opportunity for alternative processes that offer more flexibility for materials integration.

6.2 Epitaxial lift-off technology

Epitaxial lift-off (ELO) processes have been demonstrated to create free standing or bonded $\text{Al}_x\text{Ga}_{1-x}\text{As}$ epitaxial layers of low Aluminum content by sacrificially etching an $\text{Al}_x\text{Ga}_{1-x}\text{As}$ layer of higher Aluminum content in diluted hydrogen fluoride (HF) solution⁹⁷⁻¹⁰¹. A review of HF solution based ELO processes and a summary of recent developments can be found in ref [102]. The ELO approach of transferring a surface layer by lateral selective etching of an embedded layer has many advantages for engineered substrate fabrication. ELO eliminates the need for hydrogen implantation and since it can be implemented at low temperature it eliminates two of the constraints posed

by the SmartcutTM process. Combined with novel epitaxy processes and wafer bonding, ELO has the potential to fabricate challenging engineered substrate designs in a relatively cost effective manner.

6.3 Extending ELO technology with Ge/III-V heterostructures

Ge has nearly the identical lattice constant to GaAs, AlAs, and $\text{Al}_x\text{Ga}_{1-x}\text{As}$ alloys, and with the proper growth sequence can be integrated with these III-V materials to establish thin film stacks with embedded selective etch layers because Ge possesses distinct chemical properties from GaAs or AlAs. For example, Ge is chemically resistant to dilute HF solutions used in conventional ELO process. It has also been demonstrated that the noble gas halide, xenon difluoride (XeF_2), etches Si and Ge extremely selectively with respect to III-V semiconductors and insulators such as SiO_2 and SiN_x ¹⁰³⁻¹⁰⁵. Using these selective etching properties and our ability to design and fabricate high quality Ge/III-V heterostructures, we developed new approaches to extend the flexibility of ELO technology to transfer Ge or III-V epitaxial layers for fabrication of engineered substrates. It's important to note that a high level of unintentional doping in the Ge or III-V layer is acceptable for these processes, since after layer transfer additional epitaxy can be done to create the desired device structure.

6.4 Ge/AlAs/GaAs for Ge layer transfer

Figure 6.1 shows the cross-sectional TEM image of a Ge/AlAs/GaAs heterostructure. The AlAs layer and GaAs regrowth layer were grown at 650 °C. The the surface was annealed at 600 °C for 10 min to obtain a high Al-to-As ratio for Ge initiation;

subsequently, the substrate was cooled down to 450 °C for Ge epitaxy. This procedure is identical to the one for forming Ge on GaAs. High quality planar epitaxy was also observed for this Ge on AlAs case. No extended defect such as dislocation or stacking fault was observed with cross-sectional TEM imaging. The Ge surface roughness was found by atomic force microscopy (AFM) to be 2.007 nm on a $10\mu\text{m} \times 10\mu\text{m}$ scale.

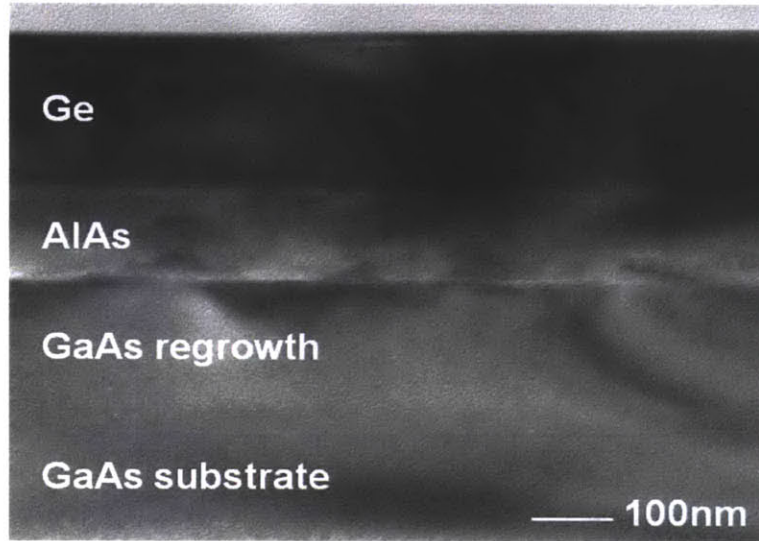


Figure 6.1. XTEM image showing the as-grown Ge/AlAs/GaAs heterostructure

This structure was used to test the feasibility of extending the conventional ELO process to transferring epitaxial Ge layers to a different substrate. A GaAs substrate with this heterostructure on top was cleaved into square pieces of $3\text{mm} \times 3\text{mm}$, or $5\text{mm} \times 5\text{mm}$. These samples were then bonded to Si substrates with a low temperature melting wax. The bonded samples were then immersed into a 10% HF solution at room temperature. The HF solution laterally etched the AlAs layer and left the Ge layer and GaAs substrate intact, allowing for the release of the bonded Ge layer from the GaAs donor substrate. The etch sequence took 72 hours and a minimal mechanical force was required to completely detach the GaAs substrate, leaving Ge bonded to the Si handle substrate. As

shown in Figure 6.2, optical microscope examination of the bonded Ge layer showed complete transfer to the Si substrate with no observable defects or etch residue. Optical microscope examination of the GaAs donor substrate surface showed no defects or etch residue as well, suggesting complete AlAs removal by lateral etching in HF solution. The RMS surface roughness of the transferred Ge layer was found by atomic force microscopy (AFM) to be 1.3 nm on a $1\mu\text{m} \times 1\mu\text{m}$ scale, which is very suitable for device application.

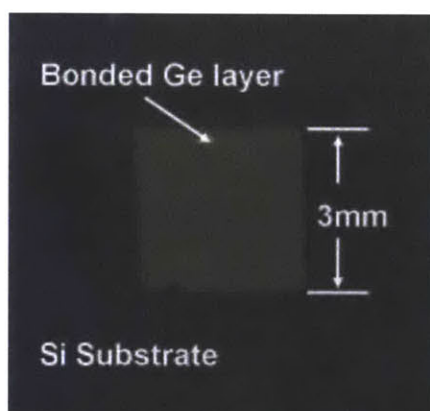


Figure 6.2. Top Ge layer shown in figure 1a transferred to a Si substrate through an epitaxial lift-off process with lateral etching of AlAs by diluted HF

The configuration of the etching apparatus made it difficult to determine the actual etch rate as a function of time or determine precisely when the AlAs layer was completely dissolved. We estimated the etch rate from samples with incomplete Ge layer transfer by measuring the undercut at various times and we calculated an etch rate of $0.02 \pm 0.01 \text{ mm/h}$ for $3\text{mm} \times 3\text{mm}$ samples. The etch rate was also found to decrease significantly with sample size, with most of the $5\text{mm} \times 5\text{mm}$ samples failing to display complete layer transfer within an acceptable time limit of 150 hours. An HF solution in 70°C wafer bath was also tested for the process; however we found that the concentration

of the solution was difficult to control due to evaporation, making it difficult to carry out the etching process in a controlled fashion. The results suggest that the etching process is limited by the ability for the acid to diffuse to the etch front through the narrow gap created by removal of the embedded AlAs layer. Further etch rate enhancing techniques, which have been demonstrated, will have to be applied to further develop this technology⁹⁹⁻¹⁰¹. Nevertheless, our results demonstrate that ELO technology using epitaxial Ge/III-V heterostructures can be used to transfer high quality epitaxial Ge layers. With further development of etch rate enhancing techniques and incorporation of HF-resistant insulating materials such as silicon nitride, this process has the potential to be used as a template to fabricate GeOI substrates.

6.5 GaAs/Ge/GaAs structure for GaAs layer transfer

Our result above again confirmed that conventional HF based ELO processes are rate limited by the ability of the HF to diffuse to the etch front through the narrow gap created by removal of the embedded layer. The etch rate can be enhanced by using mechanical force to expand the gap,^{100,101} but it is difficult to integrate a mechanical apparatus with an aqueous acid-based etching process. One way to eliminate the limit on etch rate posed by aqueous etching techniques is to employ a selective dry etch process. As mentioned above, the XeF₂ etching of Ge is a room temperature gas phase process that has been demonstrated to etch Ge with high selectivity with respect to III-V compounds, SiO₂, and SiN_x. XeF₂ spontaneously etches Ge at room temperature primarily through the following reaction:



This reaction has an enthalpy of -796 kJ per mole of Ge at 298K and occurs spontaneously at room temperature.¹⁰⁵ Both the etchant, XeF₂, and reaction products, Xe and GeF₄, are gas phase at room temperature so no etch residue remains and the reaction by-products are very mobile. Very rapid etch rates (up to 9mm/hr for α -Ge) can be achieved without the need for any plasma enhancing technique¹⁰⁴. Xuan *et al.* reported a crystalline Ge etch rate of 40 $\mu\text{m}/\text{min}$ with XeF₂ partial pressure of 0.8 Torr.¹⁰³ Previously, we published results that estimated the etch selectivity of Ge to GaAs to be on the order of 10⁶ to 1.¹⁰⁶ Additionally, gas phase etching eliminates the surface tension forces that are present in structures released with wet chemical etching. The rapid etch rates, high selectivity, and easier handling post liftoff make XeF₂-based ELO processes ideal for GaAs/Ge heterostructures.

To apply the advantageous XeF₂ etching of Ge for transfer of GaAs thin films, we prepared a heterostructure with a GaAs/Ge/GaAs epitaxial stack. The embedded Ge layer serves as the sacrificial release layer to transfer the top GaAs layer.

Reported methods for GaAs epitaxy on a Ge surface emphasize that the substrate offset and the temperature profile during growth are critical to establishing GaAs thin films on Ge that are free of anti-phase boundaries (APBs) and have a low dislocation density.⁵⁸ Incorporating that knowledge with the growth procedures of Ge on GaAs established in Chapter 3 allowed us to successfully fabricate GaAs/Ge/GaAs heterostructures that were APB-free, exhibited dislocation densities below the transmission electron microscopy (TEM) detection limit, and had an RMS surface roughness of $\sim 3\text{nm}$ on a $10\mu\text{m} \times 10\mu\text{m}$

scale. Figure 6.3 shows the cross-section TEM image of a GaAs/Ge/GaAs heterostructure and an AFM image of the surface. The surface roughness develops as a consequence of using (100) GaAs substrates offcut 6° toward the [011] direction (same as toward one of the $\langle 111 \rangle$ direction). The substrate offcut toward the [011] direction increases the [0-11] ledge density and can cause step bunches to form along [0-11] direction during growth.

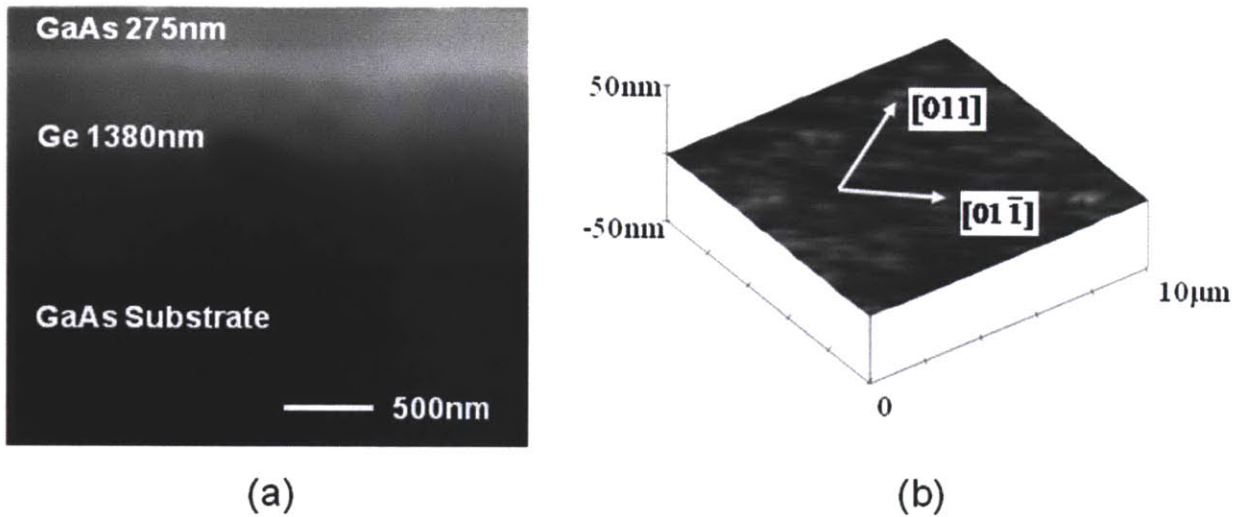


Figure 6.3. a) XTEM image of a GaAs/Ge/GaAs heterostructure grown on (100) GaAs substrate with 6° offcut toward [011] direction. No anti-phase boundary (APB) was found in the top GaAs layer. b) AFM image of the GaAs surface with formation of step bunches along [0-11] direction, leading to RMS roughness of 3.54 nm for a $10\mu\text{m}$ by $10\mu\text{m}$ scan.

To further illustrate the importance of correct substrate offcut for high quality GaAs growth on Ge, Figure 6.4 shows a XTEM image of a GaAs/Ge/GaAs heterostructure grown on (100) GaAs substrates without offcut, with exactly same surface preparation technique used for the sample shown in Figure 6.3a. According to Ting, without offcut, the Ge surface lacks double step formation, which cause the formation of APBs

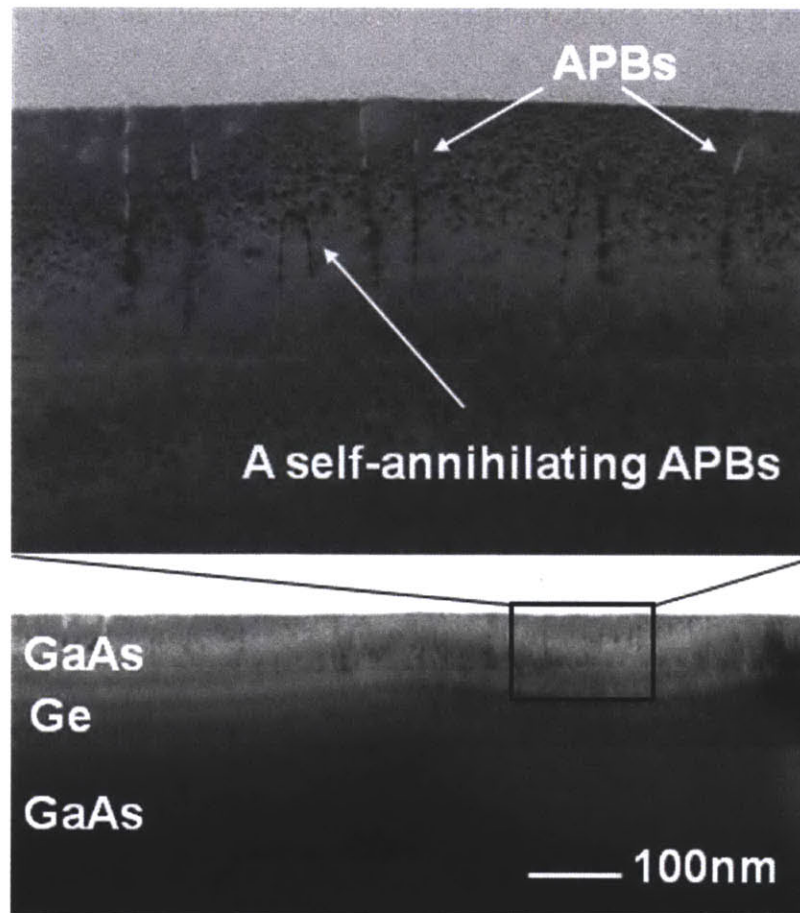


Figure 6.4. XTEM image showing APB formation in the top GaAs layer of a GaAs/Ge/GaAs heterostructure grown on (100) GaAs substrates without offcut.

As this XTEM image reveals, without offcut, a high density of APBs presented in the top GaAs layer. Although a few of them managed to annihilate each other at half the film thickness, most of them penetrate through the entire film thickness. Similar to threading dislocations, APBs generally act as scattering and recombination centers, thus would severely hinder device performance. From the TEM image, the density of APBs for this sample was estimated to be $5 \times 10^8 \text{ cm}^{-1}$, exceeding the density feasible for any electronic devices. Compared with this sample, the sample shown in Figure 6.3a grown on (100) GaAs substrates offcut 6° toward the $\langle 111 \rangle$ has no APB's or other defects within the

TEM detection limit. In the next chapter, we describe a novel low temperature process utilizing this GaAs/Ge/GaAs heterostructure to fabrication GaAsOI structures.

6.6 Summary

In this chapter we reviewed the state-of-the-art process for engineered substrate fabrication, and proposed two Ge/III-V heterostructures that have potential to enable novel cost effective process for fabricating GOI or GaAsOI substrate. The combination of the GaAs/Ge/GaAs heterostructure and XeF_2 gas phase etching of the Ge is more desirable since it is a gas-phase etching process. In the next chapter, we describe a novel low temperature process utilizing this unique combination to fabrication GaAsOI structures.

**Chapter 7 Fabrication of GaAs-on-Insulator (GaAsOI)
via Sacrificial Etching of Ge by Xenon Difluoride (XeF_2)**

7.1 Fabrication process

Figure 7.1 schematically shows our fabrication process. The process consists of four major steps: 1) epitaxial film deposition and oxide deposition, 2) wafer bonding of the donor wafer with handle wafer, and 3) XeF_2 etching to release the GaAs thin film on to the handle substrate.

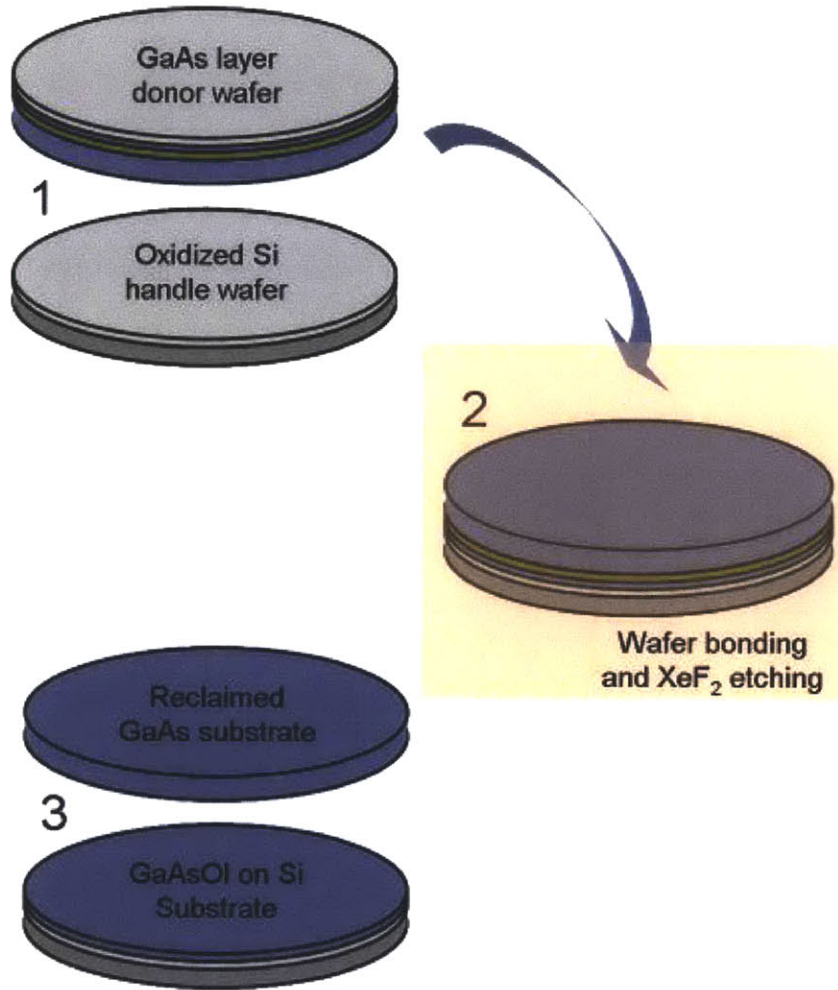


Figure 7.1. GaAsOI fabrication process in this work: 1) Preparation of the bonding substrates: thermally oxidized Si handle wafer and GaAs donor wafer with GaAs/Ge/GaAs epitaxial stack covered with PECVD oxide. Si: dark grey, SiO₂: light grey, GaAs: blue, Ge: dark yellow. 2) Low temperature wafer bonding and room temperature spontaneous etching with XeF_2 gas removing sacrificial Ge layer. 3) After separation, GaAsOI on Si substrate and reclaimed GaAs donor substrate.

First the GaAs/Ge/GaAs heterostructures described in section 6.5 were grown on epi-ready (100) GaAs substrates offcut 6° toward the [011] direction. After epitaxy the donor wafer was coated with a silicon dioxide film, which was deposited with an STS plasma-enhanced chemical vapor deposition (PECVD) system. The handle substrate was either a thermally oxidized Si substrate or a quartz substrate. Chemical mechanical polishing (CMP) was used to reduce the PECVD and quartz substrate surface RMS roughness to a level to be suitable for direct wafer bonding.

The donor wafer and the handle wafer were bonded at room temperature with direct oxide-oxide bonding, using an EV501 wafer bonder. Either a five minute Piranha clean or an O_2 -plasma treatment was used as the pre-bonding activation process. Silicon nitride was deposited on the back side of the GaAs donor substrate to protect the GaAs during the Piranha clean. The O_2 -plasma treatment was carried out in the STS PECVD system. Razor blade tests were performed on test pairs to measure the bond strength.

The XeF_2 lateral etching of the embedded Ge layer was carried out in a commercial SE Tech ES-2000XM XeF_2 etcher. A schematic of a similar system can be found in Ref [107]. The etching process followed a charge/etch/pump cycle sequence. The five second charging step consists of pressurizing the etching chamber with XeF_2 until the chamber pressure is approximately 3 Torr. The XeF_2 is provided by an attached source/expansion chamber with XeF_2 at a pressure of 3.5 Torr, the room temperature vapor pressure of XeF_2 . The source/expansion chamber pressure is maintained by the room temperature sublimation process of a XeF_2 solid source. During the etch step, XeF_2 gas in the etch

chamber etches any exposed Ge. The duration of etch step was set to 30 seconds in our process. After the etch step, the etch chamber is evacuated to less than 20 mTorr, removing the unreacted XeF₂ and reaction products and preparing the chamber for the next cycle. Throughout the process a stainless steel disc above the sample rotates at 10 rpm to enhance gas circulation. By the end of the etch sequence the Ge sacrificial layer is consumed and the donor and the handle substrate are separated, leaving the epitaxial GaAs layer bonded to the handle substrate, forming a GaAsOI stack.

7.2 Advantage of low temperature process

In standard Smartcut™ technology, the hydrogen exfoliation process requires the bonded wafer pair to be annealed at elevated temperature. This requirement makes it difficult to use Smartcut™ technology for layer transfer from a donor wafer that has a very different coefficient of thermal expansion (CTE), α , from that of the handle substrate. For instance if the bonded wafer pair were to consist of Si($\alpha=2.6\times10^{-6}/^{\circ}\text{C}$) and GaAs($\alpha=5.73\times10^{-6}/^{\circ}\text{C}$), at elevated temperatures, the strain energy per unit area from the thermal expansion mismatch would exceed the bond strength and the wafers would debond or break.^{95,96} In our process, since the XeF₂ sacrificial etching is conducted at room temperature, we eliminate the problems associated with thermal expansion mismatch between two very thick substrates. The typical oxide-oxide bond strength after room temperature direct bonding is between 50 to 200 mJ/m²,^{108,109} which is sufficient for wafer pair handling prior to and during the following etch sequence. Once the thin GaAs layer is transferred to the handle substrate, the CTE mismatch is of less concern since, the strain energy from thermal mismatch, which scales with thickness, would be much smaller than the bond

strength. Thus, the final GaAsOI structure can be annealed at high temperature to strengthen the oxide-oxide bond for subsequent device fabrication processes.

7.3 PECVD oxide layer stress control

As described in the last chapter, the GaAs/Ge/GaAs structure grown on 6° offcut substrate has a high density of step bunches. The resultant surface morphology is not suitable for direct bonding to another surface since direct bonding in general requires the surface roughness be less than 0.5 nm.^{108,109} To overcome this obstacle, PECVD oxide was deposited on the surface and CMP was used to smooth the oxide surface to an RMS roughness less than 0.5 nm. We observed that internal strain energy in the PECVD oxide can exceed the oxide-oxide bond strength and lead to film delamination. Figure 7.2 shows scanning electron microscopy (SEM) image of a GaAs/SiO₂ film stack transferred to a Si handle substrate that has buckled in a zigzag pattern, due to the high intrinsic compressive stress in the PECVD oxide layer.

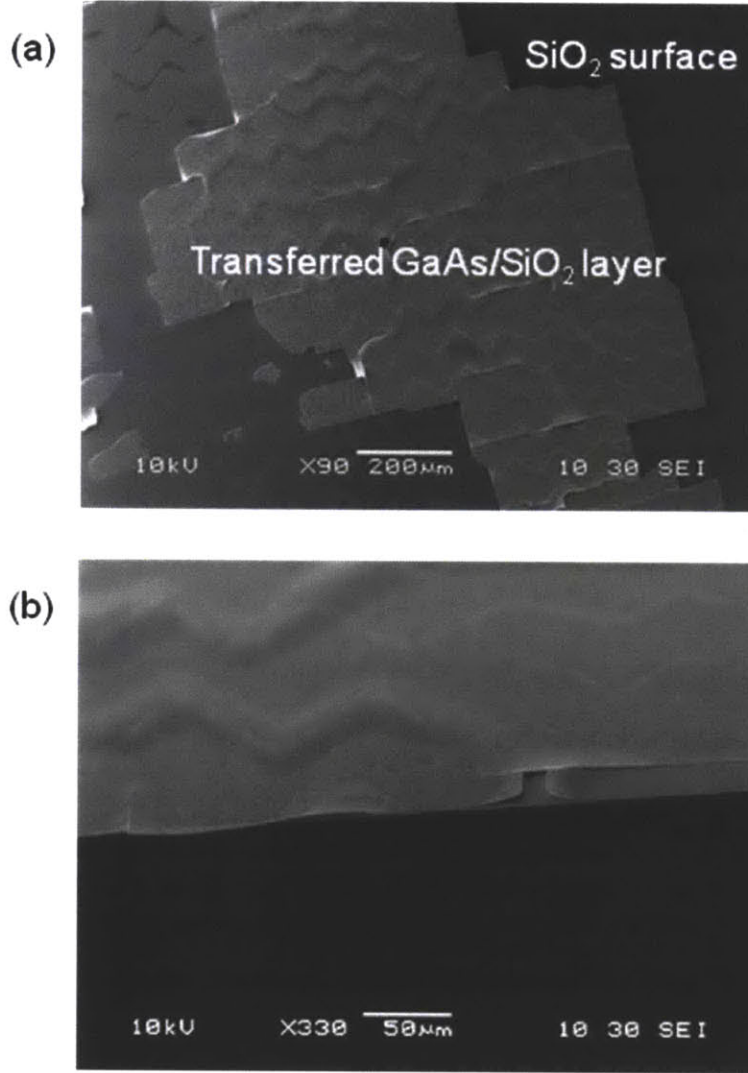


Figure 7.2. SEM images showing buckled GaAs/SiO₂ layer after being transferred to the handle substrate, due to strong intrinsic compressive stress in the PECVD oxide film. a) Top-down view of the transferred film stack. Part of the transferred film peeled off exposing the oxide surface on the handle wafer. b) 37° tilted view of the edge of the piece showing the buckled part of GaAs/SiO₂ layer detached from the handle substrate.

The strain energy per unit area, E_s , caused by PECVD oxide intrinsic stress can be calculated with the following equation:

$$E_s = \int_0^{h_s} \frac{\sigma_s^2(z)}{M_s} dz + \int_0^{h_f} \frac{\sigma_f^2(z)}{M_f} dz$$

Eq. 7.1

Here h_s , σ_s , and M_s are the thickness, stress, and biaxial modulus of the handle substrate, while h_f , σ_f , and M_f are the corresponding parameters of the bonded oxide film. The GaAs film is much thinner than the oxide film and has much less intrinsic stress, thus is not considered here. The strain level in the substrate is much lower than in the oxide film, allowing us to neglect first term in the equation. The failure condition is then determined by the balance between the strain energy in the PECVD oxide layer and the oxide-oxide bond strength.

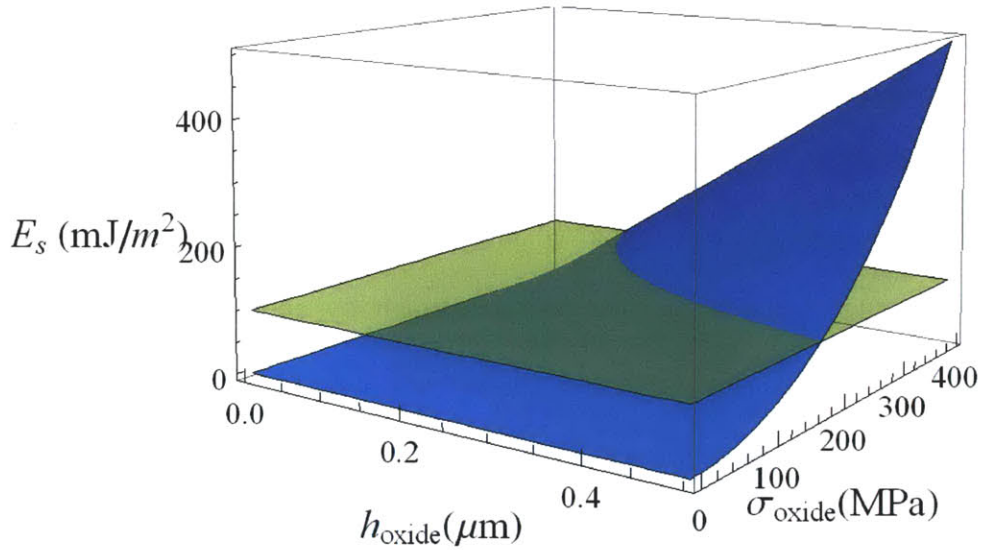


Figure 7.3. Comparison of strain energy in the PECVD oxide layer (plotted against layer thickness, h , and intrinsic stress, σ , blue plane) and the oxide-oxide bond strength (green plane).

As illustrated in Figure 7.3, taking the biaxial modulus of PECVD oxide as a constant ($\sim 80 \text{ GPa}$), the strain energy in the oxide layer scales linearly with layer thickness, and parabolically with its intrinsic stress. The oxide-oxide bond strength is a constant (setting to 100 mJ/m^2 in the plot), depending on the room temperature direct bonding condition. When the strain energy is larger than the bond strength (represented by the section of the blue plane that is above the green plane in Figure 7.3), the bond would be expected to

fail, whereas when the strain energy is less than the bond strength, the GaAs/SiO₂ stack would be expected to stay intact.

From this analysis one can see that reducing the total strain energy in the PECVD oxide through reducing thickness or intrinsic stress, as well as improving the oxide-oxide bond strength would increase the likelihood of successful GaAs layer transfer. Typically PECVD oxide experiences a compressive stress when deposited on Si or GaAs substrate due to thermal expansion mismatch. With process optimization we were able to reduce the compressive stress to 250 MPa. The oxide thickness must be a minimum of 100 nm prior to the CMP process in order for the CMP process to remove enough thickness to reduce the surface roughness to the level suitable for direct oxide-oxide bonding. One can also improve the room temperature bond strength by applying a plasma activation process.¹¹⁰⁻¹¹² For a GaAs/quartz bonded pair with both bonding surfaces treated with O₂-plasma prior to bonding, we measured bond strength of ~70 mJ/m² after room direct bonding. A low temperature anneal between 60 to 80 °C for 14 hours further improved the bonded strength to 240±20 mJ/m². At higher temperature the wafer pair would debond and annealing for greater than 14 hours showed no further improvement in bond strength. With a combination of these methods we effectively engineered our process to ensure that the bonded GaAs/SiO₂ stack survived after the XeF₂ etch sequence. Figure 7.4 shows a cross-sectional TEM image of the final GaAsOI/Si structure fabricated with this process. The oxide/oxide bond interface maintained its integrity after our process and can be further strengthened with a 800 to 1000 °C annealing step to make the GaAsOI/Si structure more robust for device fabrication processes.

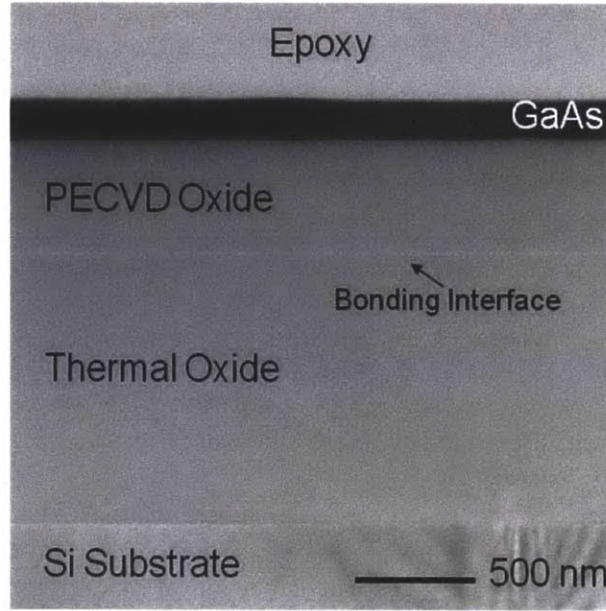


Figure 7.4. XTEM image showing GaAsOI/Si structure with high quality bond interface obtained with our low temperature fabrication process.

7.4 A model for Ge sacrificial etching

We demonstrated the feasibility of our low temperature GaAsOI fabrication process with small area pieces, but scaling the process to large diameter engineered substrates would require fast lateral etching of the Ge in order to minimize total process time. We now discuss the rate limiting mechanisms in our process and propose methods to make it more scalable.

The Deal-Grove (D-G) model for oxidation of Silicon has been widely adopted for modeling etch rate for sacrificial etching of oxides in HF ^{113,114} and photoresists in acetone.¹¹⁵ Brazzle *et al.* first adapted the D-G model to describe XeF_2 etching of Silicon.¹⁰⁷ The original D-G oxidation model states that oxide growth rate, $\dot{h}_{ox}$, can be modeled by the following equation:

$$\dot{h}_{ox} = \frac{C^*/C}{\left(\frac{1}{k_{ox}} + \frac{h_{ox}}{D_{ox}} \right)} \quad \text{Eq. 7.2}$$

Here h_{ox} is the thickness of the growing oxide. k_{ox} is the rate of constant of the oxidation reaction at the oxide/Si interface. D_{ox} is the diffusivity of the oxidant in the oxide, C^* is the concentration of the oxidant in the oxide at the oxide/air interface, and C is the molar amount of oxidant required for a unit volume of oxide. From Eq. 7.2, one can see that the oxidation rate is limited by the slower of two mechanisms: (1) the oxidation reaction happening at the oxide/Si interface or (2) the diffusion of oxidant through the oxide to the oxide/Si interface. At steady state of oxidation, all the parameters on the right side of the equation are constant except h_{ox} . As h_{ox} increases, the oxidation rate, $\dot{h}_{ox}$, decreases. In other words, oxidation process slows down as it progresses. The nature of sacrificial etching is very similar to the oxidation process. At the beginning of the process, the etch rate is limited by the reaction. As the etch progresses, transport of the etchant and etch products to and from the etch front through the narrow channel created by removal of the sacrificial layer eventually limits the process. XeF_2 has a mean free path of 10 μm at the average etching pressure of 3 Torr and since a reasonable Ge sacrificial layer thickness is normally much less than 10 μm , the XeF_2 transport to the etch front is in molecular flow regime. Brazzle *et al.* modeled this process as gas transport through a rectangular channel with width b , height c , and length L , and extracted an effective diffusivity D_{eff} , described with the following relation

$$D_{eff} = k_l \left(\frac{b \cdot c}{b + c} \right) \quad \text{Eq. 7.3}$$

where k_l is mean molecular velocity for any gas at a given temperature. For XeF_2 k_l is 139 m/s at 298K.

With D_{eff} , we can express the sacrificial etch rate, $\dot{L}$, with a formula that is analogous the oxidation rate expression in Eq. 7.2.

$$\dot{L} = \frac{P}{RT} \frac{1}{\left(\frac{1}{k_e} + \frac{L}{D_{eff}} \right) C} \quad \text{Eq. 7.4}$$

Here L is the lateral etching distance, which is also the length of the channel. P is the XeF_2 vapor pressure at the open end of the channel, which we can consider to be the chamber pressure in our case. R is the ideal gas constant. T is the temperature. k_e is the rate constant for the etching reaction, C is the number of moles of XeF_2 needed to remove one unit volume of Ge, which is equal to 0.1462 mol/cm³.

The P/RT term is the equivalent concentration of XeF_2 at the open end of the channel. The etch process can be executed in a continuous flow mode, where a XeF_2 is injected into the chamber or in a cycled fashion as described in the experimental section, where the XeF_2 concentration is not replenished continuously. For the continuous flow mode, the partial pressure of XeF_2 is constant, whereas operating in a cycled fashion leads to a decrease in XeF_2 partial pressure as the reaction proceeds. For our experiments the etch step was considered complete before the XeF_2 partial pressure dropped by 20 percent, so as a fair approximation we can consider P/RT term to be constant.

With fixed values for P/RT and C , the lateral etch rate relies solely on $1/k_e + L/D_{eff}$. At the initial stages of the etch process, L is very small so $1/k_e \gg L/D_{eff}$, so the lateral etch rate is a constant and the etch process is in the surface-reaction-limited regime. As the process progress, L increases and the L/D_{eff} term eventually becomes substantial. In the limit that $L/D_{eff} \gg 1/k_e$ and the lateral etching process is limited by gas transport through the channel, and Eq. 7.4 can be rewritten as:

$$\dot{L} = \frac{P}{RT} \cdot \frac{1}{C} \cdot \frac{D_{eff}}{L} \quad \text{Eq. 7.5}$$

Translating our model to the geometry of an actual large area layer transfer process via Ge sacrificial layer etchings, the height of the channel is determined by the embedded Ge layer thickness, c , and the width of the channel, b , is generally much larger than the Ge sacrificial layer thickness. With these considerations Eq. 4 can be simplified to

$$D_{eff} = k_l \cdot c \quad \text{Eq. 7.6}$$

and

Eq. 7.5 can be rewritten as

$$\dot{L} = \frac{P}{RT} \cdot \frac{k_l}{C} \cdot \frac{c}{L} \quad \text{Eq. 7.7}$$

Examining Eq. 7.7, we can see that for a given chamber pressure and temperature, the lateral etch rate is inversely proportional to lateral etch distance, which is the problem when considering lateral etching processes for integration of large diameter wafers.

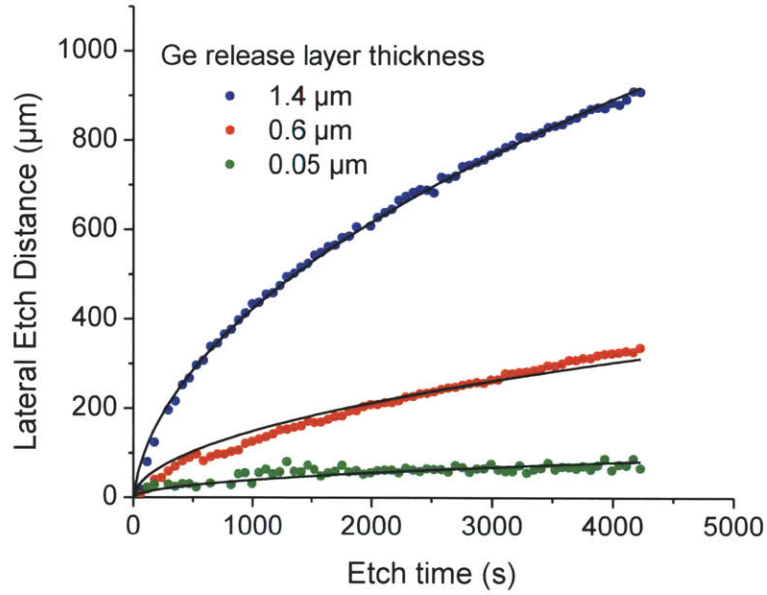


Figure 7.5. Lateral etch distance vs. etching time for three samples with different Ge layer thickness. The black solid lines are fitted curve using Eq. 7.10.

Figure 7.5 shows the measured lateral etch distance as a function of time for three samples with different Ge layer thicknesses. As can be seen in the plot the lateral etch process progresses much slower for thinner Ge layers, as predicted by Eq. 7.7. Integrating both side of this expression with respect to the etching time gives the following equation,

$$\frac{(L_f^2 - L_i^2)}{R_Q} + \frac{(L_f - L_i)}{R_L} = t_f - t_i \quad \text{Eq. 7.8}$$

where L_f and L_i are the final and initial etch front positions, at the final and initial times, t_f and t_i respectively. Parameters R_L and R_Q are the linear and quadratic rate constants, with units of $\mu\text{m/s}$ and $\mu\text{m}^2/\text{s}$, and can be expressed as the following:

$$R_L = \frac{P \cdot k_e}{R \cdot T \cdot C}; \quad R_Q = \frac{2 \cdot P \cdot k_l \cdot c}{R \cdot T \cdot C} \quad \text{Eq. 7.9}$$

Taking t_i and L_i as zero we solve Eq. 7.8 for L_f to yield the following equation.

$$L_f = k_l \cdot c \cdot \left(\frac{-1}{k_e} + \sqrt{\left(\frac{1}{k_e} \right)^2 + \frac{P}{R \cdot T} \cdot \frac{2 \cdot t_f}{C \cdot k_l \cdot c}} \right) \quad \text{Eq. 7.10}$$

Eq. 7.10 allows us to calculate the lateral etch distance for a given etch duration using only physically relevant parameters. Additionally, this expression allows for simple fitting of the measured lateral etch distance data. As discussed above, among the parameters in Eq. 7.10, P/RT and C are constants, and c was fixed by TEM measurements. Only the mean molecular velocity, k_l and etch reaction rate constant, k_e are not directly measurable and are thus chosen to be fitting parameters.

As illustrated in Figure 7.5, the results show that the data can be fitted well with curves generated according to Eq. 7.10. The fitted value of k_e is 3.26 m/s, and k_l is 118.9, 29.5, 23.7 m/s for the samples with 1.4, 0.6, 0.05 μm Ge layers respectively. k_e at 3.26 m/s suggest a linear etch rate of 2.2 $\mu\text{m/s}$, which is consistent with our measured bulk etch rate. k_l of 118.9 m/s for the 1.4 μm sample is close to the theoretical value 139 m/s for XeF_2 . We believe the reduced mean molecular velocity for samples with thinner Ge layer is due to increased probability of collisions with the channel walls, resulting in a decrease in their kinetic energy. These fitting results prove that our modified D-G model for lateral etch rate as discussed above is robust.

With the fitted values of k_e and k_l , we can plot the decrease of lateral etch rate as the etch process progresses for the three samples (Figure 7.6). Consistent with the model, lateral etch rate decreases much more dramatically for the samples with a thinner Ge sacrificial layer. Extrapolating from the fitted dataset we estimate that an etching time of more than

100 days is required to release a full 150-mm diameter wafer, which is not a practical option for the transfer of an epitaxial film. Therefore we must consider modifications to our process in order to make it viable for the fabrication of commercially relevant engineered substrates.

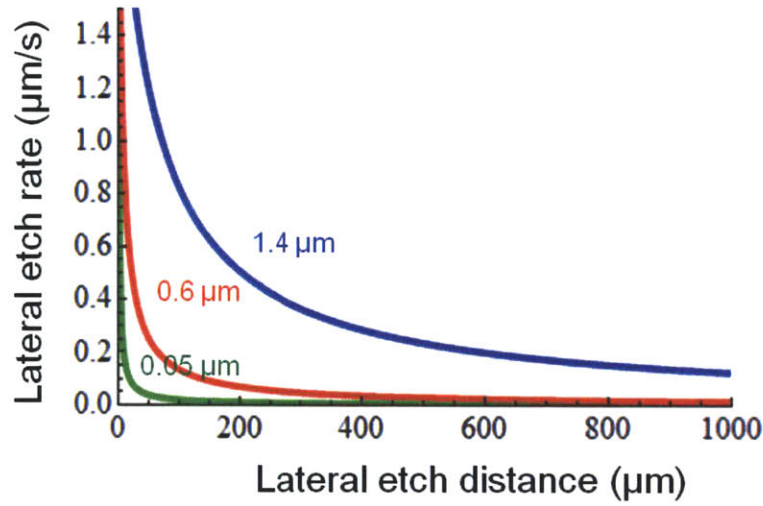


Figure 7.6. Plot of lateral etch rate vs. lateral etch distance for the three samples presented in Figure 7.5, using the parameter extracted from data fitting.

7.5 *Potential methods to extend lateral etch limit*

Our previous analysis assumed that the channel height was fixed at the value of the Ge film thickness. In practice, the channel height does not have to coincide with the Ge layer thickness. As illustrated in Figure 7.7a, the released section of the substrate sandwich can bend away from the bonded interface to create an additional displacement of δ , creating a total channel height of $c + \delta$ at the open end. This means the effective diffusivity varies along lateral etch direction. At the open end of the channel the diffusion would be much less limited, although close to the etch front effective diffusivity is still limited by c .

Overall gas transport through the channel would be enhanced and the lateral etch rate would increase.

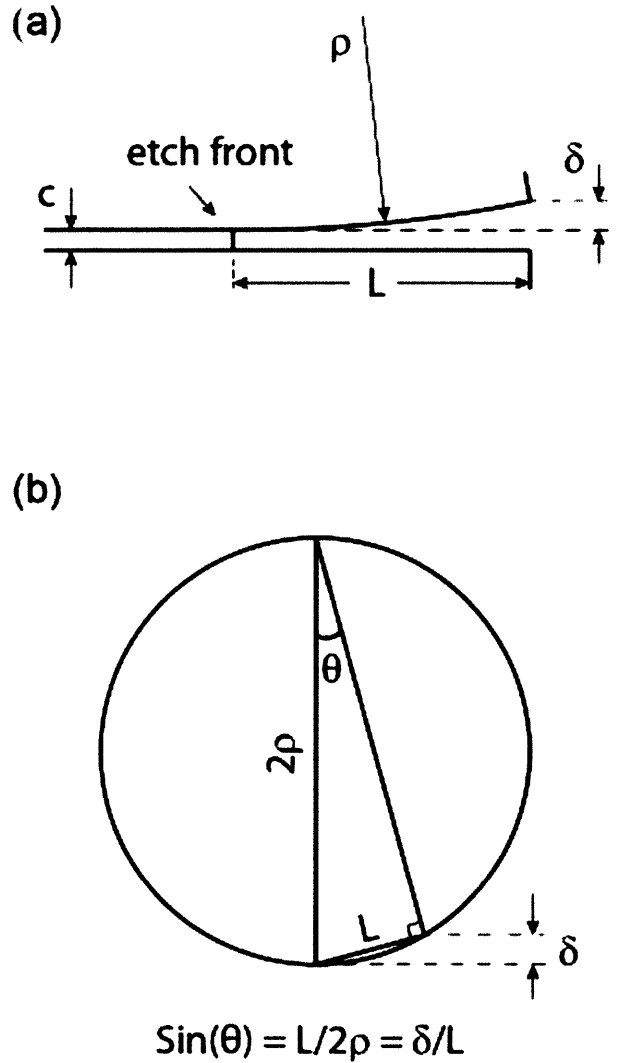


Figure 7.7. a) An illustration of gap opening during the lateral etch process b) Geometry showing the relation between L , ρ , and δ .

This concept has been applied to the development of HF solution based ELO processes when attempting to transfer large area devices such as solar cells. Several different methods to expand the channel have been demonstrated to significantly increase lateral etch rate. The first method developed to enhance lateral etch rate in HF based ELO

processes was the use of a high stress wax layer to cause the released section of the thin epitaxial film to curve up.⁹⁸ This method is generally limited to thin released layers due to the relatively low stress imparted by the wax. Weight induced separation was then developed. In this method, the donor substrate is mounted facing down onto a flexible carrier substrate. The whole stack is then mounted to the bottom of a supporting rod and submerged in an HF solution with a weight attached to one side of the flexible carrier. As the etch progresses the weight pulls one end of the flexible carrier downward, opening the gap between the donor substrate and the flexible carrier, and thus enhancing the lateral etch rate. Recently, Schermer *et al.* further increased the lateral etch rate by employing a support cylinder to apply a constant curvature to the flexible carrier.^{100,102}

We now consider the details of how expanding the channel increases lateral etch rate. As illustrated in Figure 7.7, with the additional displacement the channel height can now be expressed as $a(c+\delta)$, with a being a pre-factor that accounts for the tapered channel geometry change. The resulting formula for lateral etch rate is now given by

$$\dot{L} = a \cdot \frac{P}{RT} \cdot \frac{k_l}{C} \cdot \frac{(c + \delta)}{L} \quad \text{Eq. 7.11}$$

In any practical implementation of a method to expand the channel, the release layer thickness, c , would be much smaller than δ , so we will assume $\delta \gg c$, and Eq. 7.11 can be rewritten as

$$\dot{L} = a \cdot \frac{P}{RT} \cdot \frac{k_l}{C} \cdot \frac{\delta}{L} \quad \text{Eq. 7.12}$$

This equation shows that if one were to increase δ to keep $\delta+L$ constant, the lateral etch rate would remain constant. In addition, this constant lateral etch rate is proportional to

δ/L . Furthermore, from examining the geometry illustrated in Figure 7.7b, we can see $\delta/L=L/2\rho$, where ρ is radius of curvature, and Eq. 7.12 can be rewritten as

$$\dot{L} = a \cdot \frac{P}{RT} \cdot \frac{k_l}{C} \cdot \frac{L}{2\rho} \quad \text{Eq. 7.13}$$

This equation illustrates a significant relation that if the radius of curvature of the channel was constant, the lateral etch rate would actually increase with lateral etch distance. This observation is consistent with the results of Voncken *et al.*, where they observed that the lateral etch rate increases if ρ was kept constant,¹⁰¹ and with the results of Schermer *et al.* who significantly increased the lateral etch rate over large distance by using a cylindrical mechanical apparatus to apply a fixed curvature.^{100,102}

From our analysis, it is clear that lateral sacrificial etching is naturally limited by reactant and product transport through the channel and methods to increase the channel cross-section are critical for enhancing the lateral etch rate and extending the lateral etch distance. In practice, keeping the curvature constant is easier than trying to keep δ/L constant and it is more effective in enhancing lateral etch rate. It is important to note that the demonstrated methods for increasing the channel cross-section utilized a flexible carrier substrate that made it easy to apply large curvature (small ρ). For our current process, the sacrificial layer and etch channel are sandwiched between two full thickness wafers, which makes the overall stack rigid and less amenable to large curvature. However, it is possible as we established a curvature by depositing a highly tensile-stress silicon nitride film on the back side of the donor wafer. We measured a δ value of 50 μm across the 150 mm GaAs wafer. However, during the etch process the non-stoichiometric silicon nitride film reacted with XeF_2 and reduced its ability to induce any

wafer bow. Metal films (Pt, Ni, Al, or Ta) could be used, which do not etch or roughen in the presence of XeF_2 .¹¹⁶ Prior mechanical thinning of donor and handle substrates could also allow for larger curvature. An intermediate step using a flexible handle substrate to transfer the GaAs epi-layer can also be adopted. Finally, using an apparatus to mechanically widen the channel for chemical transport is viable for our approach because of the gas phase processing.

7.6 Summary

In this chapter, we have developed a novel low temperature process for fabrication of GaAs-on-insulator (GaAsOI) structures. This process uses room temperature oxide-oxide bonding in combination with a XeF_2 lateral etching process which is selective to Ge. This GaAsOI fabrication process proved successful on a small scale, but implementation of this process on a full wafer scale was limited because the lateral Ge etch process would eventually stall due to kinetic limitations. The rate limiting step in the lateral etch process was the reduced transport properties of etchant and reactant species through the narrow gap created by the removal of the Ge release layer. We adapted the Deal-Grove oxidation model, to establish expressions to understand the relationships between lateral etch rate, lateral etch distance, release layer thickness, channel displacements, and the radius of curvature of the donor wafer. We determined that channel opening methods are critical for applying lateral sacrificial etching techniques for large-scale implementation, and we proposed some approaches to advance our methods for fabrication of full wafer scale GaAsOI engineered substrates.

Chapter 8 Summary of Results and Suggestion of Future work

8.1 Summary of results

We investigated the growth of Ge/III-V heterostructures with metalorganic chemical vapor deposition (MOCVD), studied the initiation mechanism of Ge epitaxy on III-V compound materials, analyzed the optical and electronic properties of Ge epitaxial thin films, and explored the application of Ge/III-V heterostructures for the fabrication of advanced engineered substrates.

We first studied the initiation of Ge epitaxial growth on III-V materials surfaces by employing different surface preparation conditions to alter the stoichiometry of the III-V compound surface. We further investigated the correlation between epitaxy conditions, unintentional doping and optical properties of Ge thin films grown on GaAs and AlAs templates. With these experiments, we determined that Ge epitaxy on III-V compound surfaces initiates more readily on a group III element rich surface through a bonding and exchange process with the surface group III atoms. With the development of proper initiation sequence, we have successfully demonstrated structurally high quality tensile-strained Ge thin films and quantum dots on $\text{In}_x\text{Ga}_{1-x}\text{As}$. Low growth temperatures between 350 °C and 500 °C suppressed strain relaxation. Tensile strain levels as high as 0.58% in Ge thin films and 1.37% in Ge quantum dots were achieved, which were far higher than that previously achieved values via other methods.

A III-V compound surface with high III-to-V ratio facilitates high structural quality Ge epitaxy, but also tends to lead to elevated levels of Group III element concentrations in the Ge epitaxial layer. On the other hand, Group V (As)-rich III-V compound surfaces

lead to ‘pitting’ defect morphologies during Ge epitaxy, but the surface As does not incorporate in the Ge layer. The As concentration during Ge epitaxy is mainly dependent on gas phase transport of volatile As_2 and As_4 in the growth ambient. Ge/III-V compound heteroepitaxial processes lead to the incorporation of both group III and group V elements and generate highly doped and compensated Ge epitaxial thin films.

Luminescence studies of Ge/III-V thin films stacks showed that the low growth temperatures for the Ge epitaxy inhibited the activation of the doping species and resulted in point defect concentrations that suppressed radiative recombination. A post growth annealing sequence was required to observe luminescence from the Ge indirect band gap at 20K and the Ge direct band gap transition at 300K. Additionally, an n-type doping character was required to observe luminescence from the Ge direct band gap at room temperature. To our knowledge, our results were the first observation of room temperature luminescence of the Ge direct band gap transition from Ge epitaxial thin films grown on III-V compound substrates.

Specially designed Ge/III-V heterostructures in combination with selective sacrificial etching techniques enable novel fabrication processes for low cost manufacturing of advanced substrates. We were able to grow high quality GaAs/Ge/GaAs heterostructure that when combined with a selective XeF_2 process and room temperature oxide-oxide bonding methods, enabled us to develop a novel low temperature process for fabrication of GaAs-on-insulator (GaAsOI) structures. This GaAsOI fabrication process proved successful on a small scale, but implementation of this process on a full wafer scale was

limited because the lateral Ge etch process would eventually stall due to kinetic limitations. We adapted the Deal-Grove oxidation model to establish expressions to understand the relationships between lateral etch rate, lateral etch distance, release layer thickness, channel displacements, and the radius of curvature of the donor wafer. We determined that channel opening methods are critical for applying lateral sacrificial etching techniques for large-scale implementation.

8.2 Suggestion for future work

8.2.1 Controlling the surface exchange process

During Ge epitaxy on III-V materials, the trade-off between surface morphology and ‘auto-doping’ appears fundamental in that excellent surface morphology is achieved when there is a larger affinity for like-species (Ga and Ge) but a concomitant increase in exchange is unavoidable and results in a high level of unintentional doping in the Ge epilayer. With the understanding of this exchange mechanism, one could potentially modify the epitaxial process to reduce the level of unintentional doping. For example, after the first monolayer of Ga and Ge exchange, surface modification techniques such as introducing a third species or annealing in different ambient conditions could be applied before continued Ge growth in order to alter the surface energy and potentially ‘tie-up’ or eliminate the surface Ga. Further experimentation with these procedures could potentially yield epitaxial sequences that offer better control over unintentional doping. Tensile-strained Ge thin films with desired doping profile could then be established on which carrier mobility measurements and further optical measurements can be performed to test theoretical predictions and open the path for application in novel devices and circuits.

8.2.2 Fabrication of large diameter engineered substrates

With the small-scale demonstration of the low temperature GaAsOI fabrication process, and the established model indentifying the etch rate limiting process, it is clear that methods to fabricate large diameter engineered substrates via sacrificial Ge layer etching require approaches to enhance the lateral sacrificial etch rate through mechanical means. As mentioned in Chapter 7, process modifications such as prior mechanical thinning of donor and handle substrates, using a flexible handle substrate as intermediate transfer assistance, or applying an apparatus to mechanically widen the channel, would enhance the chemical transport and reduce the total processing time. Compared to wet etching techniques, the gas-phase XeF_2 etching of Ge makes wafer handling much easier and reduces the complexity of these modifications. With further developments, Ge/III-V heterostructures would enable novel low cost fabrication processes for more variety of engineered substrates.

Appendix 1

Matlab Code for Calculating Critical Thickness

Following Matthews¹¹⁷, critical thickness of an epitaxial film can be estimated by minimizing the sum of elastic strain energy and energy from misfit dislocations. In this consideration, the energy of a dislocation is calculated by integrating the strain energy around the dislocation from its core to a cutoff radius. Following Fitzgerald's notations¹¹⁸, the energy per unit length of a misfit dislocation with mixed edge and screw characteristics is

$$E_d = \frac{Db}{2}(1 - \nu \cos^2 \alpha)[\ln(R/b) + 1]$$

$$D = \frac{G_f G_s b}{\pi(G_f + G_s)(1 - \nu)}$$

where R is the cutoff radius. The value of R is determined according to the comparison between the epitaxial film thickness h and the average spacing between the misfit dislocations S . For thin film with $h < S/2$, the misfit dislocation interacts with the top film surface and R is approximated to be equal to h . For thicker film with $h > S/2$, the misfit dislocations interact with each other and R is better approximated to be $S/2$. In reality, for low mismatched system or if the film is grown at relatively low temperature, the average spacing between misfit dislocations is much bigger than the film thickness, and we can safely assume R equals h . Under this assumption, the critical thickness can be determined by the following equation

$$h_c = \frac{D(1 - \nu \cos^2 \alpha) \left(\frac{b}{b_{eff}} \right) \left[\ln \left(\frac{h_c}{b} \right) + 1 \right]}{2Yf}$$

% Following Bulsara¹¹⁹ and Gupta¹²⁰, this script computes and plots the critical thickness for an tensile-strained Ge thin film on a In(x)Al(1-x)As substrate over a range of Indium composition.

clear all

%Ge parameters

a_Ge=5.6578; %Ge Lattice Constant

C11=1.29e12; %Ge C11

C12=4.83e11; %Ge C12

C44=6.68e11; %Ge C44

v=C12/(C11+C12); %Poisson's ratio

Y=C11+C12-(2*C12^2/C11); %Young's Modulus of (100)

G=C44-(2*C44+C12-C11)/3; %Shear Modulus

%Substrate parameters

%AlAs parameters

a_AlAs=5.6611;

C11_AlAs=1.25e12;

C12_AlAs=5.34e11;

C44_AlAs=5.42e11;

%InAs parameters

a_InAs=6.0583;

C11_InAs=8.33e11;

C12_InAs=4.53e11;

C44_InAs=3.96e11;

x_In=0:0.001:0.2; %In composition in In(x)Al(1-x)As

%Linear interpolation of materials properties substrate at x_In

a_s=a_AlAs+(a_InAs-a_AlAs)*x_In;

C11_s=C11_AlAs+(C11_InAs-C11_AlAs)*x_In;

```

C12_s=C12_AlAs+(C12_InAs-C12_AlAs)*x_In;
C44_s=C44_AlAs+(C44_InAs-C44_AlAs)*x_In;
G_s=C44_s-(2*C44_s+C12_s-C11_s)/3; %Substrate Shear Modulus

%parameters of 60 degree misfit dislocations at Ge/In(x)Al(1-x)As interface
b=sqrt(2)/2*a_Ge; %burger's vector
b_eff=b/2; %effective burger's vector
Alpha=pi/3; %Alpha is 60 degree
sq_cos_a=1/4; %cos(alpha)^2=1/4
D=G*G_s*b./(pi*(G+G_s)*(1-v)); %average shear modulus at Ge/InAlAs interface, ./
ensure per element divide
f=abs((a_s-a_Ge)/a_Ge); %mismatch

%Calculating critical thickness by giving an initial guess of 100 angstrom
hc1=0;
hc2=100;

% Loop to establish critical thickness iteratively until difference between successive
guesses is
% less than 0.1 Å.
while(abs(hc1-hc2)>0.1)
    hc1=hc2;
    hc2=D*(1-v/4).*(log(hc1/b)+1)./(Y*f); % .* and ./ ensure operation on elements of the
matrix in stead of the matrix
end

hc2;
plot (x_In, hc2)
xlabel('x in In(x)Al(1-x)As substrate')
ylabel('Tensile-strained Ge film critical thickness (angstrom)')

```

Appendix 2

Matlab Code for Calculating Elastic Strain Remaining With Certain Thickness

When a tensile-strained Ge layer is grown on top of a $\text{In}_x\text{Al}_{1-x}\text{As}$ layer above critical thickness, misfit dislocation will form at the Ge/ $\text{In}_x\text{Al}_{1-x}\text{As}$ and total misfit, f , will be partially accommodated by plastic strain, δ , also known as relaxation. The rest of f will remain as elastic strain, ε , following the relationship: $f = \delta + \varepsilon$. Following Fitzgerald, we define δ_{eq} as the relaxation if the system reaches thermodynamic equilibrium, and δ_k as the actual relaxation allowed by kinetics^{118,121}. Accordingly, we define ε_{eq} as remaining elastic strain at thermodynamic equilibrium, and ε_k as remaining elastic strain with kinetic constrains. For the case of Ge layer grown on $\text{In}_x\text{Al}_{1-x}\text{As}$, δ_{eq} and ε_{eq} are functions of only the properties of Ge and $\text{In}_x\text{Al}_{1-x}\text{As}$, with the later largely determined by Indium composition, x ; and can be calculated by minimizing the system energy. Kinetics limited relaxation δ_k , and thus ε_k , are functions of not only material properties but also growth conditions such as temperature and growth rate. They can be calculated by considering dislocation dynamics during the growth.

$$\varepsilon_{\text{eq}} = \frac{D(1 - \nu \cos^2 \alpha) \left(\frac{b}{b_{\text{eff}}} \right) \left[\ln \left(\frac{h}{b} \right) + 1 \right]}{2Yh}$$

$$\varepsilon_{\text{eq}} = \frac{D(1 - \nu \cos^2 \alpha) \left(\frac{b}{b_{\text{eff}}} \right) \ln \left[\frac{2b(f - \varepsilon)}{b_{\text{eff}}} \right]}{2Yh}$$

Actual elastic strain preserved by kinetics ε_k , can be calculated from ε_{eq} according to the following equation:

$$\varepsilon_k = f - (f - \varepsilon_{eq})(1 - e^{-At})$$

Where
$$A = \frac{G}{2G_0} \rho b v_0 \exp\left(\frac{-E_a}{kT}\right)$$

Where, G is shear modulus of Germanium. G_0 is a constant equal to 1 MPa. ρ is the threading dislocation density (TDD) in the Ge film, which can be approximated by TDD of the InAlAs layer, assuming no dislocation nucleation occur during growth of the tensile-strained Ge layer. b is the Burger's vector, v_0 is the constant for dislocation velocity, E_a is the activation energy for dislocation glide in Germanium. The value of v_0 and E_a used here are after Yonenaga *et al.*¹²²

Fundamental parameters for Ge

	Vo	Ea
Wang	0.16x10 ¹⁴ nm/s	1.1ev

% Residual elastic strain of Ge grown on In(x)Al(1-x)As, at a certain Ge thickness,
% growth temperature and growth rate.

%Part 1: Equilibrium elastic strain

clear all

% Ge parameters

a_Ge=5.6578; % Ge Lattice Constant
C11=1.29e12; % Ge C11 unit: pa
C12=4.83e11; % Ge C12
C44=6.68e11; % Ge C44

v=C12/(C11+C12); % Poisson's ratio
Y=C11+C12-(2*C12^2/C11); % Young's Modulus of (100)

```

G=C44-(2*C44+C12-C11)/3; % Shear Modulus

% Substrate parameters
% AlAs parameters
a_AlAs=5.6611;
C11_AlAs=1.25e12;
C12_AlAs=5.34e11;
C44_AlAs=5.42e11;

% InAs parameters
a_InAs=6.0583;
C11_InAs=8.33e11;
C12_InAs=4.53e11;
C44_InAs=3.96e11;

x_In=0.08; % In composition in In(x)Al(1-x)As
% Linear interpolation of materials properties substrate at x_In
a_s=a_AlAs+(a_InAs-a_AlAs)*x_In;
C11_s=C11_AlAs+(C11_InAs-C11_AlAs)*x_In;
C12_s=C12_AlAs+(C12_InAs-C12_AlAs)*x_In;
C44_s=C44_AlAs+(C44_InAs-C44_AlAs)*x_In;
G_s=C44_s-(2*C44_s+C12_s-C11_s)/3; % Substrate Shear Modulus

% parameters of 60 degree misfit dislocations at Ge/In(x)Al(1-x)As interface
b=sqrt(2)/2*a_Ge; % burger's vector
b_eff=b/2; % effective burger's vector
Alpha=pi/3; % Alpha is 60 degree
sq_cos_a=1/4; % cos(alpha)^2=1/4
D=G*G_s*b./(pi*(G+G_s)*(1-v)); % average shear modulus at Ge/InAlAs interface, ./
ensure per element divide
f=abs((a_s-a_Ge)/a_Ge); % mismatch

% Calculating critical thickness by giving an initial guess of 100 angstrom
hc1=0;
hc2=100;

% Loop to establish critical thickness iteratively until difference between successive
guesses is
% less than 0.1 Å.
while(abs(hc1-hc2)>0.1)
    hc1=hc2;
    hc2=D*(1-v/4).*(log(hc1/b)+1)./(Y*f); % .* and ./ ensure operation on elements of the
matrix in stead of the matrix
end
hc2;

```

```

h1=1:1:hc2;
h2=hc2:1:2000;

% Caculating epsilon at different h

% epsilon=f for h<hc2
epsilon1=f*ones(1,length(h1));

% caulating epsilon for hc2<h<hc4
epsilon2=D*(1-v/4).*(log(h2/b)+1)./(Y*h2);

h=[h1 h2];
epsilon_eq=[epsilon1 epsilon2];

% Part 2, calculate actual residue elastic strain limited by kinetics at a certain temperature

% Parameters of dislocation glide
Ea=1.62; % activation energy, unit: eV value from ,
T=450; % Growth temperature, C
k=8.617e-5; % Boltzmann constant, unit: eV/K
Go=1e6; % constant, unit: pa
vo=2.9e12; % dislocation volocity constant, unit A/s
rho_cm=5e5; % threading dislocation density, unit /cm^2
rho=rho_cm*1e-16; % threading dislocation density, unit /A^2
rg=0.61; % growth rate at temperature T and substrat composition x, unit: A/s
t=h/rg; % growth time, unit: s

A=(rho*b*vo*G*exp(-Ea/(k*(T+273))))/(2*Go);
epsilon_k=f-(f-epsilon_eq).*(1-exp(-A*t));
epsilon=[epsilon_eq; epsilon_k];

plot (h,epsilon*100)
xlabel('Tensile-strained Ge film thickness (angstrom)');
ylabel('Residual elastic strain (%)');
legend1=legend('\epsilon_e_q','\epsilon_k');
title(['x_I_n=',num2str(x_In),', Temperature=',num2str(T),'C,', ' Growth
Rate=',num2str(rg),'A/s,', ' TDD_s _t=',num2str(rho_cm)])

```

Appendix 3

Mathematica Script for Calculating Elastic Energy of Bonded Wafer Pair

Elastic energy with bonded pair of Si/GaAs or Si/Ge at different temperature;

	Si	Ge	GaAs
C_{11} [N/m ²]	1.66×10^{11}	1.24×10^{11}	1.19×10^{11}
C_{12} [N/m ²]	6.39×10^{10}	4.13×10^{10}	5.38×10^{10}
C_{44} [N/m ²]	7.96×10^{10}	6.83×10^{10}	5.99×10^{10}
α [K ⁻¹]	2.59×10^{-6}	5.8×10^{-6}	6.86×10^{-6}
$\gamma^{(001)}$ [J/m ²]	1.34	—	0.91

Clear[T];

Materials Parameters ;

$$C_{11, Si} = 1.66 \times 10^{11};$$

$$C_{12, Si} = 0.639 \times 10^{11};$$

$$C_{44, Si} = 0.796 \times 10^{11};$$

$$\alpha_{Si} = 2.59 \times 10^{-6};$$

$$C_{11, GaAs} = 1.19 \times 10^{11};$$

$$C_{12, GaAs} = 0.538 \times 10^{11};$$

$$C_{44, GaAs} = 0.599 \times 10^{11};$$

$$\alpha_{GaAs} = 6.86 \times 10^{-6};$$

$$C_{11, Ge} = 1.24 \times 10^{11};$$

$$C_{12, Ge} = 0.413 \times 10^{11};$$

$$C_{44, Ge} = 0.683 \times 10^{11};$$

$$\alpha_{Ge} = 5.8 \times 10^{-6};$$

Biaxial Modulus of the two wafers;

$$M_1 = C_{11, Si} + C_{12, Si} - 2C_{12, Si}^2 / C_{11, Si};$$

$$M_2 = C_{11, Ge} + C_{12, Ge} - 2C_{12, Ge}^2 / C_{11, Ge};$$

Wafer thicknesses;

$$h_1 = 0.65 \times 10^{-3};$$

$$h_2 = 0.65 \times 10^{-3};$$

Calculating bonded pair curvature at T;

$$\Delta T = T - 20;$$

$$\rho = ((M_2 h_2^3 + M_1 h_1^3)(M_2 h_2 + M_1 h_1) + 3M_2 h_2 M_1 h_1 (h_1 + h_2)^2) / (6M_2 h_2 M_1 h_1 (h_1 + h_2) (\alpha_{Si} - \alpha_{GaAs}) \Delta T);$$

Calculating stress distribution along thickness of the wafers;

$$\sigma_1 = 1/\rho ((z - h_1/2)M_1 - 2/(h_1(h_1 + h_2)) ((M_2 h_2^3 + M_1 h_1^3)/12));$$

$$\sigma_2 = 1/\rho ((z - h_2/2)M_2 - 2/(h_2(h_1 + h_2)) ((M_2 h_2^3 + M_1 h_1^3)/12));$$

Calculating total strain energy;

$$E_{el} = \int_0^{h_1} \frac{\sigma_1^2}{2M_1} dz + \int_{-h_2}^0 \frac{\sigma_2^2}{2M_2} dz;$$

Plot[1000E_{el}, {T, 20, 1000}, PlotRange → {0, 3000}, Axes → False]

Plot[ρ, {T, 20, 100}]

Appendix 4

Mathematica Script for Calculating Strain Energy in Strained Oxide Layer Bonded to Si Wafer

(Same as Strained Oxide Layer Grown on Si Wafer)

Oxide Parameters

Clear[s]

Biaxial Modulus;

$$M_{\text{oxide}} = 0.8 \times 10^{11};$$

Residue Strain;

$$\sigma_{\text{oxide,MPa}} = -250;$$

Calculating curvature from oxide strain and thickness h in microns;

$$\rho = - (1/6) \quad (M_1 \quad h_1^2 / (h_{\mu\text{m}} \times 10^{-6} \times \sigma_{\text{MPa}} \times 10^6));$$

Strain distribution in the substrate;

$$\sigma_1 = M_1 / \rho \quad (z - (2h_1) / 3) ;$$

Plot [σ_1 , {z, 0, h_1 }] ;

Strain energy in film;

$$E_f = \int_0^{h_{\mu\text{m}} \times 10^{-6}} \frac{(\sigma_{\text{MPa}} \times 10^6)^2}{2 M_{\text{oxide}}} dz ;$$

Strain energy in substrate (note: Strain in substrate is very small compared to the film strain)(strain energy in substrate is very small compared to the strain energy in the film);

$$E_s = \int_0^{h_1 \times 10^{-2}} \frac{\sigma_1^2}{2 M_1} dz ;$$

Show[Plot3D[E_{total,mJ}=1000(E_s+E_f),{h_{μm},0,0.5},{σ_{MPa},0,400},Mesh→None,AxesLabel→Automatic,PlotLabel→"Energy (mJ/m²)",TicksStyle→Directive[12],LabelStyle→Directive[Bold,16]],Plot3D[100,{h_{μm},0,0.5},{σ_{MPa},0,400},Mesh→None,PlotStyle→Directive[Yellow,Opacity[0.5]]]]

strainedoxidelayerbondedtoSiwafer

Appendix 5

Mathematica Script for Gas Phase Lateral Etch Rate

Note: This model is for gas phase lateral etch rate. Wet lateral etch follows same principle but the transport terms should be modified for liquid transport. For example, pressure term (P/RT) should be replaced with etchant concentration in liquid, and gas transport term $k_l \cdot h$ should be replaced with liquid diffusion.

Model Equations

Lateral Etch Rate:

$$\dot{L} = \frac{P}{RT} \frac{1}{\left(\frac{1}{k_e} + \frac{L}{k_l \cdot h} \right) C}$$

Lateral Etch Distance:

$$L = k_l h \left(-\frac{1}{k_e} + \sqrt{\left(\frac{1}{k_e} \right)^2 + \frac{2P}{RTC} \frac{t}{k_l \cdot h}} \right)$$

Units

L, lateral etch distance, in μm

V, lateral etch rate, in $\mu\text{m/s}$

t, time, in s

P, pressure, in Torr. 1 Torr = 133.3 Pa

R, idea gas constant. $R=8.3145 \text{ cm}^3 \text{ MPa K}^{-1} \text{ mol}^{-1}$

T, temperature, in K

P/RT, effective concentration, in mol/cm^3

Co, amount of etchant to etch unit volume of sacrificial material, in mol/cm^3

P/RTC, ratio between P/RT and C, unitless

K_l , Etchant gas molecule mean velocity, in $\mu\text{m/s}$

h, Channel height, in μm

R_L , Linear etch rate constant, in $\mu\text{m/s}$

K_e , Etch reaction constant, in $\mu\text{m/s}$, $K_e=R_L/(P/RTC)$

Input Parameters

Values for XeF_2 etching of Ge, in units above

```
Clear["Global`*"];
```

```
P=3;
```

```
R=8.3145;
```

```
T=293;
```

```
Co=0.1462;
```

```

kl=139*^6;
h={0.1,0.6,1.4};
RL=1;

```

Calculations

Lateral etch rate

```
a=P*133.3*10^-6/(R*T*Co);
```

```
V=a/(a/RL+X/(kl*h));
```

```
L=kl*h*(-a/RL+sqrt((a/RL)^2+2*a*t/(kl*h)));
```

Plots

```
Plot[V,{X,0,3000}]
```

```
Plot[L,{t,0,900}]
```

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